

χ -TWISTED LUCAS CONGRUENCES FOR THE SECOND SOLUTIONS OF APÉRY-LIKE RECURRENCES

ABSTRACT. Let $(A(n))_{n \geq 0}$ be one of the fifteen sporadic Apéry-like sequences, and let $(B(n))_{n \geq 0}$ be the second solution of its three-term recurrence, normalised by $B(0) = 0$, $B(1) = 1$. The first row satisfies a Lucas congruence, $A(ap + r) \equiv A(a)A(r) \pmod{p}$; for the second row no congruence of Lucas type appears in the literature. We prove that the second row descends along base- p digits as well, subject to two corrections: a normalising factor p^w , where w is the total degree of the sequence's harmonic weight, and a quadratic Dirichlet character χ . Concretely, for the classical Apéry pair for $\zeta(3)$ we prove

$$p^3 b_{ap+r} \equiv b_a a_r \pmod{p} \quad (p \geq 5, 1 \leq a < p, 0 \leq r < p),$$

by an elementary self-contained argument, and we then prove a general theorem: if the second solution of a sporadic pair admits a harmonic-monomial weight of total degree w satisfying five explicit hypotheses, then $p^w B(ap + r) \equiv \chi(p)^e B(a)A(r) \pmod{p}$, with $e \in \{0, 1\}$ read off the weight. The character is not decorative. The untwisted law fails completely for seven of the fifteen families, at exactly the primes where $\chi(p) = -1$, and the single factor $\chi(p)$ – which we trace to one line of Frobenius descent for a χ -twisted harmonic letter – repairs it. For the twelve families that have an archimedean Apéry limit, w and χ are its weight and character; for the other three both are read off the harmonic weight directly, and the descent holds regardless. Exact computation over all primes $5 \leq p \leq 103$ supports a master form, $v_p(p^w B_n A_q - \chi(p) B_q A_n) \geq w$ with $q = \lfloor n/p \rfloor$, with floor exactly w in every cell tested; we state it as a conjecture and never as a theorem. Three of the fifteen sequences have no archimedean Apéry limit at all, and satisfy the p -adic descent regardless. Along the way we prove a new closed form for the Apéry numerators, $b_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2 (2H_n^{(3)} - H_k^{(3)})$, which we have not located in the literature and which reduces the weight-three congruence to three lines of the general theorem. The general theorem, both first-row Lucas congruences and two instances of the second-row congruence are formalised in Lean 4, sorry-free. In the language of Beukers–Vlasenko's Dwork crystals our congruence concerns the non-unit diagonal entry of the Frobenius matrix, complementary to their Conjecture 7.5 about the unit root; the character is invisible in unit-root formulations and has to be put in by hand.

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Date: July 25, 2026.

2020 Mathematics Subject Classification. 11A07, 11B65, 11S80, 33F10.

Key words and phrases. Apéry numbers, Lucas congruences, Dwork congruences, second solution, sporadic sequences, p -adic Apéry limits, creative telescoping.

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1. INTRODUCTION

1.1. **The first row.** Let

$$a_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 = 1, 5, 73, 1445, 33001, 819005, 21460825, \dots \quad (1.1)$$

be Apéry's numbers for $\zeta(3)$ (OEIS A005259). They satisfy a congruence of Lucas type: for every prime p , every $a \geq 0$ and every r with $0 \leq r < p$,

$$a_{ap+r} \equiv a_a a_r \pmod{p}, \quad (1.2)$$

and consequently $a_n \equiv \prod_i a_{n_i} \pmod{p}$ for the base- p expansion $n = \sum_i n_i p^i$. This is classical [22], and it is by now known to be a structural, not accidental, feature of the whole *sporadic* class: Malik and Straub [13, Thm. 3.1] prove (1.2) for every one of the fifteen sporadic Apéry-like sequences in Zagier's classification, by McIntosh's constant-term method, and Straub [18] sharpens it to a congruence modulo p^2 for all fifteen. Beyond Lucas congruences there is a supercongruence: $a_{p^r m} \equiv a_{p^{r-1} m} \pmod{p^{3r}}$ for $p \geq 5$ [17, Thm. 3.2], a statement whose history runs through Beukers' conjecture and Coster's proof, as recorded in the survey of Osburn and Sahu [16]. On the crystalline side, Beukers and Vlasenko's Dwork-crystal papers [2, 3, 4] and Mellit–Vlasenko [14] put the same first-row congruences into a Frobenius framework, where they appear as statements about the *unit root* $F(t)/F(t^\sigma)$ of the holomorphic period.

In short: the arithmetic of the integral solution of an Apéry-like recurrence is well-trodden ground.

1.2. **The second row, and the gap.** Apéry's proof of the irrationality of $\zeta(3)$ uses a *second* solution of the same recurrence,

$$b_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 \left(H_n^{(3)} + \sum_{m=1}^k \frac{(-1)^{m-1}}{2m^3 \binom{n}{m} \binom{n+m}{m}} \right), \quad H_x^{(r)} = \sum_{m=1}^x \frac{1}{m^r}, \quad (1.3)$$

with $b_n/a_n \rightarrow \zeta(3)$; the first values are $0, 6, \frac{351}{4}, \frac{62531}{36}, \frac{11424695}{288}, \dots$ (OEIS A059415 for the numerators). The pair (a_n, b_n) is not integral: b_n has denominators, bounded by $d_n^3 b_n \in \mathbb{Z}$ with $d_n = \text{lcm}(1, \dots, n)$.

For this second row, nothing of the shape (1.2) appears in the literature. We surveyed the relevant technology in detail, working only from fetched sources, and the picture is uniform:

- *Lucas congruences for constant-term sequences* (Malik–Straub [13], Henningsen–Straub [12], Mellit–Vlasenko [14], Straub [17, 18]) treat the integral solution only.
- *Dwork crystals* [2, 3, 4] construct the logarithmic second solution $F(t) \log t + G(t)$, whose coefficients are exactly our $B(n)$; they explain the p^w normalisation, since the Cartier matrix on the rank-two quotient is $\Lambda = Y \Lambda_0 (Y^\sigma)^{-1}$ with $\Lambda_0 = \begin{pmatrix} \alpha_0 & \alpha_1 \\ 0 & p\alpha_0 \end{pmatrix}$, so that the second-solution Frobenius eigenvalue carries exactly one extra factor of p per Hodge step [4, Prop. 7.7]; and they prove p -integrality of the mirror map $q(t) = t \exp(G/F)$ [4, Cor. 7.11]. What they do *not* prove is any congruence modulo p or p^s for the second solution: the corresponding deepening is their Conjecture 7.5, explicitly open, and stated for the *unit root* at that.
- *Mirror-map integrality* (Delaygue–Rivoal–Roques [10], Delaygue [9], [15]) and *strong Frobenius structures* (Vargas-Montoya [20, 19]) control the denominators of the second-solution ratio, and in [19] prove Lucas congruences — but explicitly only for the *holomorphic* solution.
- *p -adic zeta values* sit in the Frobenius matrix entries of a MUM operator (Beukers–Vlasenko [5]), which is the conceptual reason a weight- w second solution should descend with a factor p^w — but that is a statement about a *p -adic limit*, proved for simplicial and hyperoctahedral Calabi–Yau families, and the classical Apéry operator is not among them.

- The one genuine precedent is Gessel’s companion congruence, in the form given by Beukers [1, Thm. 1.4]: with $A'_k = \sum_{m=0}^k \binom{k}{m}^2 \binom{m+k}{m}^2 \left(\frac{1}{m+1} + \cdots + \frac{1}{k}\right)$ one has $A_{kp+\ell} \equiv A_\ell A_k + p A'_\ell k A_k \pmod{p^2}$. This is a *weight-one* harmonic companion, and the depth stops at p^2 . (We have confirmed the modulus p^2 and the definition of A' against the source; the placement of the indices in the p -term should be checked against the published version before it is relied upon.)

So the second row is a genuine gap: weight one is known modulo p^2 ; weight three, the weight of $\zeta(3)$ and of Apéry’s own companion, is not known at all.

1.3. Results.

(A) *The weight-three theorem.* Our first result is elementary and self-contained.

Theorem 1.1 (weight-three descent; Theorem 2.18 below). *Let $p \geq 5$ be prime, let $1 \leq a < p$ and $0 \leq r < p$, and put $n = ap + r$. Then b_a and $p^3 b_n$ lie in $\mathbb{Z}_{(p)}$ and*

$$p^3 b_{ap+r} \equiv b_a a_r \pmod{p}.$$

The proof occupies Section 2; it needs Lucas’ theorem, Kummer’s carry criterion, and nothing else. The three steps that a referee found compressed in the first write-up — the borrow count in $\binom{n}{m}$, the product-completion of the summation region, and the interchange defining the tail sums — are here isolated as Lemma 2.6, Lemma 2.10 and Lemma 2.1 respectively, and proved in full.

(B) *The master depth law.* Numerically the congruence of Theorem 1.1 is the mod- p shadow of a sharper and more symmetric statement which we can verify but not prove. Writing $q = \lfloor n/p \rfloor$, the quantity $v_p(p^3 b_n a_q - b_q a_n)$ is ≥ 3 , with minimum exactly 3, for every prime $5 \leq p \leq 31$ and every $n \leq 320$ — including the multi-digit range and including $p = 5$. Section 3 states this as Conjecture 3.1 and records the exact sweep parameters, the fine structure of the sporadic boosts above the floor, and two false statements that the data kills. The descent depth equals the motivic weight of the Apéry limit; this is what makes the programme’s name for it, *the master depth law*, appropriate.

(C) *The general χ -twisted theorem.* The classification of Apéry-like recurrences produces fifteen sporadic pairs $(A(n), B(n))$. Running the same experiment on all fifteen produces the paper’s main discovery: the naive law is *false* for seven of them, and the failure is governed by a quadratic Dirichlet character.

Conjecture 1.2 (master form; Conjecture 3.6 below). *For each sporadic pair, with w the weight of its Apéry limit, χ the quadratic character attached to it, $p \geq 5$ and $q = \lfloor n/p \rfloor$,*

$$v_p(p^w B_n A_q - \chi(p) B_q A_n) \geq w.$$

[VERIFIED: zero failures] for all fifteen pairs, all primes $5 \leq p \leq 103$, all $n \leq 400$, and $n \leq 1200$ for $p = 5, 7, 11$ (five base- p digits), in exact rational arithmetic; the floor is exactly w in every single (sequence, prime) cell.

The mod- p , single-digit form of Conjecture 1.2 we can prove, in a generality that covers four of the fifteen families outright:

Theorem 1.3 (Theorem 4.4 below). *Let $p \geq 5$ and let $B(n) = \sum_k S(n, k) w(n, k)$, where S is a product of binomials in linear forms and w is a $\mathbb{Q}$ -combination of weight- w monomials in harmonic letters $K_\chi^{(r)}(x) = \sum_{m \leq x} \chi(m) m^{-r}$. Under the five hypotheses (H1)–(H5) of Section 4,*

$$p^w B(ap + r) \equiv \chi(p)^e B(a) A(r) \pmod{p} \quad (1 \leq a < p, 0 \leq r < p),$$

where $e \in \{0, 1\}$ is the common number of χ -letters per monomial.

The mechanism is a two-line lemma (Lemma 4.1): for a completely multiplicative χ ,

$$p^r K_\chi^{(r)}(y) = \chi(p) K_\chi^{(r)}(\lfloor y/p \rfloor) + p^r \cdot (p\text{-integral}),$$

because the p -divisible indices $m = jp$ contribute $\chi(jp)(jp)^{-r} = \chi(p)p^{-r}\chi(j)j^{-r}$. The factor $\chi(p)$ in the congruence is exactly the $\chi(p)$ coming out of the p -divisible layer of the harmonic weight. Correspondingly, χ is the character of the sequence's Apéry limit: the limit is a rational multiple of $\zeta(w)$ when χ is trivial, and of $L(\chi_D, w)$ when $\chi = \chi_D$. The correlation is 12 out of 12 on the sequences whose limit exists.

(D) *A minimal closed form.* The general theorem needs a harmonic-monomial weight, and finding one for Apéry's b_n produced a formula we have not been able to locate anywhere:

$$b_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 (2H_n^{(3)} - H_k^{(3)}). \quad (1.4)$$

Section 6 proves (1.4) by six explicit Zeilberger/Gosper certificates, and proves two further closed forms of the same kind, for Franel's numbers and for $\sum_k \binom{n}{k}^4$. Substituted into Theorem 1.3, (1.4) reduces Theorem 1.1 to three lines. It also removes the hypothesis $p \geq 5$ for that instance, since the coefficients 2, -1 are integers.

(E) *Sharpness, positioning, formalisation.* Section 7 documents the failure of the master law at $p = 2, 3$, and locates the obstruction where the proof says it should be: in the denominators of the weight's coefficients. Section 8 positions Conjecture 1.2 against Beukers–Vlasenko's Conjecture 7.5: the two are statements about the two diagonal entries of one 2×2 Frobenius matrix, neither implies the other, and the character is invisible in the unit-root formulation. Section 9 reports what is machine-verified: Theorem 1.3, Lemma 4.1, the two first-row Lucas congruences, and two instances of the second-row congruence are formalised in Lean 4, sorry-free, on the three standard Mathlib axioms.

1.4. Evidence classes. This paper reports a mixture of proofs and of exact finite computations, and it labels them. We use exactly three markers, and we never upgrade one to another:

[PROVED]:

a complete proof is written out here.

[VERIFIED]:

an exact finite computation, always with its range stated. Exact rational or modular arithmetic, no floating point. This is evidence and never proof, however large the range.

[OPEN]:

neither.

Statements carrying only [VERIFIED] are called *conjectures*, and are stated in `conjecture` environments, with the sweep parameters attached. Where a computation was run twice, by independent implementations, we say so.

1.5. Notation. Throughout, p is a prime, v_p is the p -adic valuation on $\mathbb{Q}$ normalised by $v_p(p) = 1$, and $\mathbb{Z}_{(p)} = \{x \in \mathbb{Q} : v_p(x) \geq 0\}$ is the local ring; a congruence $x \equiv y \pmod{p^s}$ between rationals means $v_p(x - y) \geq s$. We write $d_n = \text{lcm}(1, \dots, n)$, $H_x^{(r)} = \sum_{m=1}^x m^{-r}$ (so $H_0^{(r)} = 0$), and for a completely multiplicative arithmetic function χ ,

$$K_\chi^{(r)}(y) = \sum_{m=1}^y \frac{\chi(m)}{m^r},$$

which for $\chi \equiv 1$ is $H_y^{(r)}$. For a sporadic pair we write $A(n)$ for the integral solution and $B(n)$ for the second solution normalised by $B(0) = 0$, $B(1) = 1$; for the classical $\zeta(3)$ pair we write a_n and b_n , with the relation $b_n = 6B(n)$. Digits: for $n = ap + r$ with $0 \leq r < p$ we call $a = \lfloor n/p \rfloor$ the high digit and r the low digit, and similarly $k = bp + s$. Finally $\binom{n}{k} = 0$ whenever $k > n$ or $k < 0$, which we use without comment.

2. THE WEIGHT-THREE THEOREM

This section is self-contained. Its only inputs are Lucas' theorem on binomial coefficients modulo p and Kummer's carry criterion for $v_p\binom{N}{M}$.

2.1. The pair, and the harmonic decomposition. Write

$$A(n, k) = \binom{n}{k}^2 \binom{n+k}{k}^2, \quad u(n, m) = \frac{(-1)^{m-1}}{2m^3 \binom{n}{m} \binom{n+m}{m}}, \quad c(n, k) = H_n^{(3)} + \sum_{m=1}^k u(n, m), \quad (2.1)$$

so that $a_n = \sum_{k=0}^n A(n, k)$ and $b_n = \sum_{k=0}^n A(n, k) c(n, k)$ as in (1.1), (1.3). Since $H_n^{(3)}$ does not depend on k , we may split off the constant letter:

$$b_n = H_n^{(3)} a_n + W(n), \quad W(n) := \sum_{k=0}^n A(n, k) \sum_{m=1}^k u(n, m). \quad (2.2)$$

Both terms of (2.2) will be treated separately, and the theorem is the sum of the two congruences (2.7) and (2.9) below.

We shall also need the tail sums of the summand,

$$T(n, m) := \sum_{k=m}^n A(n, k) \in \mathbb{Z} \quad (0 \leq m \leq n+1), \quad (2.3)$$

with the convention $T(n, m) = 0$ for $m > n$.

Lemma 2.1 (interchange). *For every $n \geq 0$,*

$$W(n) = \sum_{m=1}^n u(n, m) T(n, m).$$

Proof. All sums are finite, so we may reorder freely. The $k = 0$ term of $W(n)$ has an empty inner sum and contributes 0; the remaining terms are indexed by the triangular region $\{(m, k) : 1 \leq m \leq k \leq n\}$, over which

$$W(n) = \sum_{k=1}^n \sum_{m=1}^k A(n, k) u(n, m) = \sum_{1 \leq m \leq k \leq n} A(n, k) u(n, m) = \sum_{m=1}^n u(n, m) \sum_{k=m}^n A(n, k),$$

which is the assertion. Note that every $u(n, m)$ occurring has $1 \leq m \leq n$, so $\binom{n}{m} \neq 0$ and $\binom{n+m}{m} \neq 0$ and no division by zero occurs. $\square$

Remark 2.2. Lemma 2.1 is one of the three steps flagged as compressed in the referee pass of the first write-up; it is elementary, but it is the definition of the object $T(n, m)$ on which the whole W -part rests, so we have stated it as a lemma.

2.2. Lucas and Kummer inputs.

Lemma 2.3 (Lucas step). *Let p be prime, $a, b \geq 0$ and $0 \leq r, s < p$. Then $\binom{ap+r}{bp+s} \equiv \binom{a}{b} \binom{r}{s} \pmod{p}$.*

Proof. The base- p expansion of $ap+r$ has last digit r and higher digits those of a ; that of $bp+s$ has last digit s and higher digits those of b . Lucas' theorem gives $\binom{ap+r}{bp+s} \equiv \binom{r}{s} \prod_{j \geq 1} \binom{a_{j-1}}{b_{j-1}}$ where a_j, b_j are the digits of a, b ; and $\prod_{j \geq 1} \binom{a_{j-1}}{b_{j-1}} \equiv \binom{a}{b}$ by Lucas' theorem again. $\square$

Lemma 2.4 (carry annihilation). *Fix $a, b \geq 0$ and $0 \leq r, s < p$, and put $n = ap + r$, $k = bp + s$. Then*

$$\binom{n+k}{k} \equiv \begin{cases} \binom{a+b}{b} \binom{r+s}{s} & \text{if } r+s < p, \\ 0 & \text{if } r+s \geq p, \end{cases} \pmod{p}.$$

Proof. We have $n+k = (a+b)p + (r+s)$. If $r+s < p$ this is a legitimate digit decomposition and Lemma 2.3 applies directly. If $r+s \geq p$, write $n+k = (a+b+1)p + (r+s-p)$ with $0 \leq r+s-p < p$; Lemma 2.3 then gives $\binom{n+k}{k} \equiv \binom{a+b+1}{b} \binom{r+s-p}{s}$, and $r < p$ forces $r+s-p < s$, so $\binom{r+s-p}{s} = 0$. $\square$

Lemma 2.5 (summand factorisation). *With $n = ap + r$, $k = bp + s$ as above,*

$$A(n, k) \equiv [r+s < p] \cdot \binom{a}{b}^2 \binom{a+b}{b}^2 \cdot \binom{r}{s}^2 \binom{r+s}{s}^2 \pmod{p},$$

where $[\cdot]$ is the indicator of a condition. In particular $A(n, k) \equiv 0 \pmod{p}$ whenever $s > r$ or $r+s \geq p$.

Proof. $A(n, k) = \binom{n}{k}^2 \binom{n+k}{k}^2$. By Lemma 2.3, $\binom{n}{k}^2 \equiv \binom{a}{b}^2 \binom{r}{s}^2$; by Lemma 2.4, $\binom{n+k}{k}^2 \equiv [r+s < p] \binom{a+b}{b}^2 \binom{r+s}{s}^2$. Multiply. The last sentence follows since $\binom{r}{s} = 0$ for $s > r$. $\square$

The next two lemmas are the entire Kummer ledger, and they are where the hypothesis $a < p$, i.e. $n < p^2$, is used.

Lemma 2.6 (borrow count). *Let p be prime, $n = ap + r$ with $0 \leq a, r < p$, and let $0 \leq m \leq n$ have digits $m = m_1p + m_0$, $0 \leq m_0 < p$. Then*

$$v_p \binom{n}{m} = [m_0 > r] \in \{0, 1\}.$$

Proof. By Kummer's theorem, $v_p \binom{n}{m}$ is the number of borrows in the base- p subtraction $n - m$. Since $n < p^2$ and $m \leq n$, both n and m have at most two base- p digits, so there are at most two borrow positions, and $m_1 \leq a$.

At position 0 a borrow occurs precisely when $m_0 > r$. At position 1 a borrow occurs precisely when $m_1 + [m_0 > r] > a$. We check that this never happens.

Case $m_0 \leq r$. Then position 1 requires $m_1 > a$, contradicting $m_1 \leq a$.

Case $m_0 > r$. We claim $m_1 \leq a - 1$. Indeed, if $m_1 \geq a$ then $m = m_1p + m_0 \geq ap + m_0 > ap + r = n$, contradicting $m \leq n$. Hence $m_1 + 1 \leq a$ and there is no borrow at position 1.

There is no position ≥ 2 , so the borrow count is $[m_0 > r]$. $\square$

Remark 2.7. The step " $m_0 > r \Rightarrow m_1 \leq a - 1$ " is the second of the three compressions flagged by the referee pass; in the first write-up it was abbreviated to "the second borrow would need $m_1 = a$ with $m_0 > r$, i.e. $m > n$ ".

Lemma 2.8 (carry count). *With $n = ap + r$, $0 \leq a, r < p$, and $0 \leq m = m_1p + m_0 \leq n$, put*

$$c_0 = [r + m_0 \geq p], \quad c_1 = [a + m_1 + c_0 \geq p].$$

Then $v_p\binom{n+m}{m} = c_0 + c_1 \leq 2$.

Proof. By Kummer's theorem this valuation is the number of carries in the base- p addition $n + m$. The digits of n are (a, r) and those of m are (m_1, m_0) , all in $[0, p)$. A carry out of position 0 occurs iff $r + m_0 \geq p$, i.e. iff $c_0 = 1$; a carry out of position 1 occurs iff $a + m_1 + c_0 \geq p$, i.e. iff $c_1 = 1$. At position 2 the digits are 0 and 0 and the incoming carry is $c_1 \leq 1 < p$, so no further carry is produced. $\square$

Lemma 2.9 (level- a carry). *If $0 \leq b \leq a < p$ and $a + b \geq p$ then $v_p\binom{a+b}{b} \geq 1$, hence $v_p A(a, b) \geq 2$. Consequently, if $0 \leq j \leq a < p$ and $a + j \geq p$ then*

$$v_p T(a, j) \geq 2.$$

Proof. $a, b < p$ and $a + b \geq p$ means the base- p addition of a and b carries out of position 0, so $v_p\binom{a+b}{b} \geq 1$ by Kummer, and $A(a, b) = \binom{a}{b}^2 \binom{a+b}{b}^2$ has valuation at least 2. In $T(a, j) = \sum_{b=j}^a A(a, b)$ every index satisfies $b \geq j$, hence $a + b \geq a + j \geq p$, so every summand has valuation ≥ 2 . $\square$

2.3. The fibre lemma, and the first row. The following lemma does the combinatorial work three times over: it yields the first-row Lucas congruence, and it yields the two “tail-factorisation” facts needed for the second row. Its proof contains the product-completion argument in full; this is the third of the compressions flagged in the referee pass, and it is the one that genuinely needs care, because the true summation region is *not* a product set.

Lemma 2.10 (fibre lemma). *Let p be prime, $n = ap + r$ with $a \geq 0$ and $0 \leq r < p$. For $0 \leq \beta \leq a + 1$ set*

$$T_\beta(n) := \sum_{\substack{0 \leq k \leq n \\ [k/p] \geq \beta}} A(n, k).$$

Then

$$T_\beta(n) \equiv T(a, \beta) \cdot a_r \pmod{p}, \quad T(a, \beta) = \sum_{b=\beta}^a A(a, b).$$

Proof. Write each k in the range uniquely as $k = bp + s$ with $0 \leq s < p$; then $b = [k/p]$ and $0 \leq k \leq n = ap + r < (a + 1)p$ forces $\beta \leq b \leq a$. By Lemma 2.5,

$$T_\beta(n) \equiv \sum_{b=\beta}^a \sum_{\substack{0 \leq s < p \\ r+s < p \\ bp+s \leq n}} \binom{a}{b}^2 \binom{a+b}{b}^2 \binom{r}{s}^2 \binom{r+s}{s}^2 \pmod{p}. \quad (2.4)$$

The constraint $bp + s \leq n = ap + r$ is vacuous for $b < a$, and for $b = a$ it reads $s \leq r$. Now, all summands with $b = a$ and $s > r$ carry the factor $\binom{r}{s}^2 = 0$ and therefore vanish *identically*, not merely modulo p . Hence they may be adjoined to the sum (2.4) without changing it, and after this adjunction the index region is the product set

$$\{\beta \leq b \leq a\} \times \{0 \leq s < p : r + s < p\},$$

in which the two constraints involve b and s separately. The double sum therefore factors:

$$T_\beta(n) \equiv \left(\sum_{b=\beta}^a \binom{a}{b}^2 \binom{a+b}{b}^2 \right) \left(\sum_{\substack{0 \leq s < p \\ r+s < p}} \binom{r}{s}^2 \binom{r+s}{s}^2 \right) \pmod{p}. \quad (2.5)$$

The first factor is $T(a, \beta)$ — an equality of integers, by the definition (2.3) of T applied at level a .

For the second factor we claim

$$\sum_{\substack{0 \leq s < p \\ r+s < p}} \binom{r}{s}^2 \binom{r+s}{s}^2 \equiv a_r \pmod{p}. \quad (2.6)$$

Indeed $a_r = \sum_{s=0}^r \binom{r}{s}^2 \binom{r+s}{s}^2$. Terms with $s > r$ have $\binom{r}{s} = 0$, so the range may be taken to be $0 \leq s < p$. It remains to see that the terms with $s \leq r$ and $r+s \geq p$ vanish modulo p : for those, $\binom{r+s}{s} = \binom{1 \cdot p + (r+s-p)}{0 \cdot p + s} \equiv \binom{1}{0} \binom{r+s-p}{s} = 0$ by Lemma 2.3, because $0 \leq r+s-p < s$. This proves (2.6) and, with (2.5), the lemma. $\square$

Theorem 2.11 (first row; Lucas congruence for a_n). **[PROVED]** *Let p be prime. For every $a \geq 0$ and every $0 \leq r < p$,*

$$a_{ap+r} \equiv a_a a_r \pmod{p},$$

and hence $a_n \equiv \prod_i a_{n_i} \pmod{p}$ for the base- p expansion $n = \sum_i n_i p^i$.

Proof. Take $\beta = 0$ in Lemma 2.10: the left-hand side is $T_0(n) = \sum_{k=0}^n A(n, k) = a_n$ and the right-hand side is $T(a, 0) a_r = a_a a_r$. The product form follows by induction on the number of base- p digits of a , the two-digit statement being available for every $a \geq 0$. $\square$

Theorem 2.11 is classical [22, 13, 18]; we have reproved it because the fibre lemma it rests on is used twice more below, and because the whole point of this paper is what happens one row down.

Corollary 2.12 (tail factorisation). *Let $n = ap + r$ with $a \geq 0$, $0 \leq r < p$, and let $1 \leq m \leq n$ have digits $m = m_1 p + m_0$.*

- (i) *If $m_0 > r$ then $T(n, m) \equiv T(a, m_1 + 1) a_r \pmod{p}$.*
- (ii) *If $m_0 = 0$, say $m = jp$ with $1 \leq j \leq a$, then $T(n, jp) \equiv T(a, j) a_r \pmod{p}$.*

Proof. (ii) The condition $k \geq jp$ is exactly $\lfloor k/p \rfloor \geq j$, so $T(n, jp) = T_j(n)$ and Lemma 2.10 applies verbatim.

(i) Here $T(n, m)$ and $T_{m_1+1}(n)$ differ only by the terms with $m \leq k < (m_1 + 1)p$, i.e. by those k with $\lfloor k/p \rfloor = m_1$ and low digit $s = k - m_1 p \geq m_0 > r$. By the last sentence of Lemma 2.5, each such term is $\equiv 0 \pmod{p}$. Hence $T(n, m) \equiv T_{m_1+1}(n) \equiv T(a, m_1 + 1) a_r \pmod{p}$. $\square$

2.4. The constant letter.

Lemma 2.13 (H -part). *Let $p \geq 3$, $n = ap + r$ with $0 \leq a, r < p$. Then*

$$p^3 H_n^{(3)} \equiv H_a^{(3)} \pmod{p^3},$$

and consequently

$$p^3 H_n^{(3)} a_n \equiv H_a^{(3)} a_a a_r \pmod{p}. \quad (2.7)$$

Proof. The indices $m \leq n$ divisible by p are $m = jp$ for $1 \leq j \leq \lfloor n/p \rfloor = a$, and they contribute $\sum_{j=1}^a (jp)^{-3} = p^{-3} H_a^{(3)}$. Writing $S = \sum_{m \leq n, p \nmid m} m^{-3} \in \mathbb{Z}_{(p)}$ we get $H_n^{(3)} = S + p^{-3} H_a^{(3)}$, whence $p^3 H_n^{(3)} = p^3 S + H_a^{(3)} \equiv H_a^{(3)} \pmod{p^3}$. Multiplying by the integer a_n preserves the congruence modulo p^3 , so $p^3 H_n^{(3)} a_n \equiv H_a^{(3)} a_n \pmod{p^3}$, in particular modulo p . Finally $H_a^{(3)} \in \mathbb{Z}_{(p)}$ because $a < p$, and $a_n \equiv a_a a_r \pmod{p}$ by Theorem 2.11; multiplying the latter congruence by $H_a^{(3)} \in \mathbb{Z}_{(p)}$ gives (2.7). $\square$

2.5. The singular layer. The terms of $W(n)$ indexed by m with $p \nmid m$ can carry a pole of order up to 3, and it is exactly the perfect-square shape of $A(n, k)$ that keeps them under control after multiplication by p^3 .

Lemma 2.14 (singular layer). *Let $p \geq 5$, $n = ap + r$ with $1 \leq a < p$, $0 \leq r < p$. For every m with $1 \leq m \leq n$ and $p \nmid m$,*

$$v_p(p^3 u(n, m) T(n, m)) \geq 1.$$

Consequently
$$\sum_{1 \leq m \leq n, p \nmid m} p^3 u(n, m) T(n, m) \equiv 0 \pmod{p}.$$

Proof. Since $p \geq 5$ and $p \nmid m$ we have $v_p(2m^3) = 0$, so with $CC := \binom{n}{m} \binom{n+m}{m}$,

$$v_p(p^3 u(n, m) T(n, m)) = 3 - v_p(CC) + v_p T(n, m). \quad (2.8)$$

By Lemmas 2.6 and 2.8, $v_p(CC) = [m_0 > r] + c_0 + c_1 \leq 3$.

If $v_p(CC) \leq 2$ then (2.8) is $\geq 3 - 2 + 0 = 1$, since $T(n, m)$ is an integer.

If $v_p(CC) = 3$ then necessarily $[m_0 > r] = c_0 = c_1 = 1$. From $m_0 > r$ and $c_1 = 1$, i.e. $a + m_1 + 1 \geq p$, we get: Corollary 2.12(i) applies, so $T(n, m) \equiv T(a, m_1 + 1) a_r \pmod{p}$; and $m_1 + 1 \geq p - a$, so every index $b \geq m_1 + 1$ occurring in $T(a, m_1 + 1)$ satisfies $a + b \geq a + m_1 + 1 \geq p$, whence $v_p A(a, b) \geq 2$ by Lemma 2.9 and $T(a, m_1 + 1) \equiv 0 \pmod{p}$. Therefore $v_p T(n, m) \geq 1$ and (2.8) is $\geq 3 - 3 + 1 = 1$. $\square$

Remark 2.15. Lemma 2.14 is the load-bearing step, and the only place where the *square* shape of $A(n, k) = \binom{n}{k}^2 \binom{n+k}{k}^2$ is essential: a single Kummer carry buys $v_p \geq 2$, which is precisely the slack that rescues the case $v_p(CC) = 3$. A summand with an unsquared wide binomial would buy only $v_p \geq 1$ and the ledger would not close. This is a concrete obstruction for any port of the argument to a non-square summand.

2.6. The p -divisible layer.

Lemma 2.16 (W -part). *Let $p \geq 5$, $n = ap + r$ with $1 \leq a < p$, $0 \leq r < p$. Then*

$$p^3 W(n) \equiv a_r W(a) \pmod{p}. \quad (2.9)$$

Proof. By Lemma 2.1 and Lemma 2.14,

$$p^3 W(n) \equiv \sum_{j=1}^a \tau_j^{(n)} \pmod{p}, \quad \tau_j^{(n)} := p^3 u(n, jp) T(n, jp) = \frac{(-1)^{j-1} T(n, jp)}{2j^3 \binom{n}{jp} \binom{n+jp}{jp}},$$

where we used $p^3/(jp)^3 = j^{-3}$ and $(-1)^{jp-1} = (-1)^{j-1}$ (p odd). Likewise, by Lemma 2.1 at level a ,

$$W(a) = \sum_{j=1}^a \tau_j, \quad \tau_j := u(a, j) T(a, j) = \frac{(-1)^{j-1} T(a, j)}{2j^3 \binom{a}{j} \binom{a+j}{j}}.$$

We compare $\tau_j^{(n)}$ with $a_r \tau_j$ for each j separately. Throughout, Lemma 2.3 gives

$$\binom{n}{jp} = \binom{ap+r}{jp+0} \equiv \binom{a}{j} \binom{r}{0} = \binom{a}{j}, \quad \binom{n+jp}{jp} = \binom{(a+j)p+r}{jp+0} \equiv \binom{a+j}{j} \pmod{p}, \quad (2.10)$$

the second because $n+jp = (a+j)p+r$ with $0 \leq r < p$ is a legitimate digit decomposition, Lemma 2.3 being valid for arbitrary high digits. Note also $v_p\binom{n}{jp} = 0$ by Lemma 2.6 (here $m_0 = 0 \leq r$) and $v_p\binom{n+jp}{jp} = c_0 + c_1 = 0 + [a+j \geq p]$ by Lemma 2.8, while $v_p\binom{a}{j} = 0$ and $v_p\binom{a+j}{j} = [a+j \geq p]$ since $j \leq a < p$.

Case $a+j < p$. Then all four binomials in sight are p -units, and by (2.10) the two at level n are congruent modulo p to the two at level a ; a p -unit congruence may be inverted modulo p . Combining with Corollary 2.12(ii), $T(n, jp) \equiv T(a, j)a_r$, we obtain

$$\tau_j^{(n)} \equiv \frac{(-1)^{j-1} T(a, j) a_r}{2j^3 \binom{a}{j} \binom{a+j}{j}} = a_r \tau_j \pmod{p}.$$

Case $a+j \geq p$. We show that both $\tau_j^{(n)}$ and τ_j are $\equiv 0 \pmod{p}$. For τ_j : $v_p \tau_j = v_p T(a, j) - v_p \binom{a}{j} - v_p \binom{a+j}{j} \geq 2 - 0 - 1 = 1$ by Lemma 2.9. For $\tau_j^{(n)}$: we have $v_p \tau_j^{(n)} = v_p T(n, jp) - 0 - 1$, and we claim $v_p T(n, jp) \geq 2$. Indeed every k in the range of $T(n, jp) = \sum_{k \geq jp} A(n, k)$ has $b := \lfloor k/p \rfloor \geq j$, so with $s = k - bp$ the carry count of Lemma 2.8 for $\binom{n+k}{k}$ satisfies $c_1 = [a+b+c_0 \geq p] \geq [a+j \geq p] = 1$, whence $v_p \binom{n+k}{k} \geq 1$ and $v_p A(n, k) \geq 2$. Therefore $v_p \tau_j^{(n)} \geq 2 - 1 = 1$.

Summing over j and discarding the terms with $a+j \geq p$ from both sides,

$$p^3 W(n) \equiv \sum_{j=1}^a \tau_j^{(n)} \equiv \sum_{\substack{1 \leq j \leq a \\ a+j < p}} a_r \tau_j \equiv a_r \sum_{j=1}^a \tau_j = a_r W(a) \pmod{p}. \quad \square$$

2.7. The theorem.

Lemma 2.17 (local integrality). *Let $p \geq 5$ and $1 \leq a < p$, $0 \leq r < p$, $n = ap + r$. Then $b_a \in \mathbb{Z}_{(p)}$ and $p^3 b_n \in \mathbb{Z}_{(p)}$.*

Proof. For $b_a = H_a^{(3)} a_a + W(a)$: the letter $H_a^{(3)}$ is p -integral because $a < p$, and by Lemma 2.1 $W(a) = \sum_{j=1}^a u(a, j) T(a, j)$, whose terms have $v_p(u(a, j) T(a, j)) = -[a+j \geq p] + v_p T(a, j)$, which is ≥ 0 when $a+j < p$ and $\geq -1 + 2 = 1$ when $a+j \geq p$ by Lemma 2.9.

For $p^3 b_n = p^3 H_n^{(3)} a_n + p^3 W(n)$: the first term is p -integral by Lemma 2.13. For the second, use Lemma 2.1 again and split the range of m . The terms with $p \nmid m$ have $v_p(p^3 u(n, m) T(n, m)) \geq 1$ by Lemma 2.14. The terms with $m = jp$ have $v_p(p^3 u(n, jp) T(n, jp)) \geq 0$: this is the content of the two cases $a+j < p$ and $a+j \geq p$ established inside the proof of Lemma 2.16, where the first gives $v_p \geq 0$ directly and the second gains the two powers of Lemma 2.9 against a pole of order at most 1. $\square$

Theorem 2.18 (second row, weight three). [PROVED] *Let $p \geq 5$ be prime, $1 \leq a < p$, $0 \leq r < p$ and $n = ap + r$. Then both sides below lie in $\mathbb{Z}_{(p)}$ and*

$$p^3 b_{ap+r} \equiv b_a a_r \pmod{p}.$$

Proof. Local integrality is Lemma 2.17. By (2.2), $p^3 b_n = p^3 H_n^{(3)} a_n + p^3 W(n)$, and by (2.7) and (2.9),

$$p^3 b_n \equiv H_a^{(3)} a_a a_r + a_r W(a) = a_r (H_a^{(3)} a_a + W(a)) = a_r b_a \pmod{p}. \quad \square$$

Remark 2.19 ($p = 5$ is not exceptional). The proof uses only that p is odd, that $p \nmid 2$, and that $1 \leq j \leq a < p$ implies $p \nmid j$; it therefore applies verbatim at $p = 5$. This matters, because at $p = 5$ one has $a_1 = 5 \equiv 0$ and $a_3 = 1445 \equiv 0 \pmod{5}$, so the *ratio* form $p^3(b_n/a_n) \equiv b_a/a_a$ acquires a pole (of order as low as -1) at the cells where a base- p digit of a meets a zero of $a_\bullet$. Theorem 2.18 never divides by a_a , so it sees no pole; the pole is an artefact of dividing, not a defect of the congruence. Zeros of $a_\bullet$ modulo p occur for $r \in \{1, 3\}$ ($p = 5$), $\{5\}$ ($p = 11$), $\{3, 13\}$ ($p = 17$), $\{8, 10\}$ ($p = 19$), $\{8, 22\}$ ($p = 31$), and for none of r when $p = 7, 13, 23, 29$ [VERIFIED: $5 \leq p \leq 31$].

Remark 2.20 (independent verification). Theorem 2.11 and Theorem 2.18, together with every lemma above and every case split in their proofs, were checked by an independent exact-arithmetic implementation — one not derived from the write-up — with zero failures: the theorems for $p \in \{5, 7, 11, 13, 17\}$ and all cells (a, r) ; the fibre and tail factorisations for $p \in \{5, 7, 11\}$ and all (a, r, m) ; the Kummer counts of Lemmas 2.6 and 2.8 for $p \in \{5, 7, 11, 13\}$ and all $p \nmid m$; and the ledger of Lemma 2.14, whose minimum over all $p \nmid m$ is exactly 1, so the bound is sharp. The dangerous branch $v_p(CC) = 3$ was confirmed on all 14 292 of its occurrences for $p \leq 19$, and $v_p(CC)$ was never observed to exceed 3.

Remark 2.21 (what is *not* proved here). Theorem 2.18 is a single-digit, modulo- p statement. The multi-digit master form and the sharpening from modulo p to modulo p^3 are [VERIFIED] only; see Section 3. The single-digit case is the base case of a digit-by-digit induction, but the induction step is not available, because the natural inductive statement is the ratio form and the ratio form has poles (Remark 2.19).

3. THE MASTER DEPTH LAW

Theorem 2.18 is a modulo- p statement about a single base- p digit. The numerical evidence says that the true statement is stronger in three independent directions at once: it holds modulo p^3 , it holds for all n (not only $n < p^2$), and it needs no p -unit hypothesis anywhere. We state it as a conjecture and we do not upgrade it.

3.1. The weight-three master form. For $n \geq 1$ put $q = \lfloor n/p \rfloor$ and

$$E(n) := v_p(p^3 b_n a_q - b_q a_n). \quad (3.1)$$

This quantity is defined for every n and every p — no division occurs, so the $p = 5$ and exceptional-digit cells are not special.

Conjecture 3.1 (master depth law, weight three). *For every prime $p \geq 5$ and every $n \geq 1$, with $q = \lfloor n/p \rfloor$,*

$$p^3 b_n a_q \equiv b_q a_n \pmod{p^3}, \quad \text{i.e.} \quad E(n) \geq 3.$$

Evidence. [VERIFIED: zero failures; all primes $5 \leq p \leq 31$; all $n \in [1, 320]$; exact rational arithmetic]. This range covers the single-digit cells $n < p^2$, the multi-digit range, and $p = 5$. The floor is attained: $\min_n E(n) = 3$ exactly, for every one of the nine primes. On p -unit cells the statement is equivalent to the ratio descent

$$p^3 \frac{b_n}{a_n} \equiv \frac{b_q}{a_q} \pmod{p^3},$$

which iterates down the whole base- p expansion; and the minimum of $v_p(p^3 b_n/a_n - b_q/a_q)$ over p -unit single-digit cells is likewise exactly 3, for every prime $7 \leq p \leq 31$, with no cell of depth ≤ 2 [VERIFIED: $7 \leq p \leq 31$].

Restricted to single-digit cells $n = ap + r$ ($1 \leq a < p$, $0 \leq r < p$) the quantity (3.1) is $E(a, r) = v_p(p^3 b_n a_a - b_a a_n)$, and its distribution is remarkably rigid: overwhelmingly 3, with a handful of sporadic boosts.

TABLE 1. Distribution of the single-digit depth $E(a, r) = v_p(p^3 b_{ap+r} a_a - b_a a_{ap+r})$ over all cells $1 \leq a < p$, $0 \leq r < p$. [VERIFIED: exact; zero cells with $E < 3$].

p	$\min E$	E -value tally over single-digit cells (E : count)
5	3	3:18, 4:1, 6:1
7	3	3:35, 4:5, 5:1, 6:1
11	3	3:104, 4:4, 6:2
13	3	3:147, 4:7, 5:1, 6:1
17	3	3:260, 4:10, 5:1, 6:1
19	3	3:302, 4:35, 5:4, 6:1
23	3	3:478, 4:25, 5:2, 6:1
29	3	3:774, 4:35, 5:2, 6:1
31	3	3:912, 4:16, 5:1, 6:1

3.2. Fine structure of the boosts. The cells with $E > 3$ are not noise, but they do not follow a clean law either. They concentrate at extreme digits, and the interior ones are symmetric under the reflection $(a, r) \mapsto (p-1-a, p-1-r)$ — a symmetry which says nothing about the $a = p-1$ boosts, since it carries them to $a = 0$, outside the range of the table [VERIFIED: $5 \leq p \leq 31$]:

- $E = 6$ at the corner $(a, r) = (p-1, p-1)$, i.e. $n = p^2 - 1$, at every prime in range;
- $E \in \{4, 5, 6\}$ at the edge $(a, r) = (p-1, 0)$, i.e. $n = p(p-1)$: it is 5 at $p = 7, 13, 17, 19, 23, 29, 31$, but 4 at $p = 5$ and 6 at $p = 11$. At $p = 11$ the edge therefore carries a second $E = 6$ cell, which is the whole of that prime's 6:2 entry in Table 1;
- $E = 4$ on symmetric interior and edge pairs, e.g. for $p = 13$ at $(3, 5)$, $(3, 7)$, $(9, 5)$, $(9, 7)$, $(12, 2)$, $(12, 6)$, $(12, 10)$.

The robust, conjecturable content is the floor $E \geq 3$; the boosts we record as data.

3.3. Two strengthenings that the data kills. It is worth recording two natural-looking sharpenings that are *false*, because both were conjectured at some point in this programme.

Observation 3.2. The floor is *flat*: it is independent of $\kappa = v_p\binom{2n}{n}$. Cells with $\kappa = 0, 1, 2$ all floor at $E = 3$ [VERIFIED: $5 \leq p \leq 31$, $n \leq 320$]. In particular the bound $E \geq v_p(a_a a_n) + 1$ is false: $E - v_p(a_a a_n)$ reaches -2 at $p = 17$.

Observation 3.3. The integrality statement $v_p(b_n) \geq -3v_p(d_n)$, equivalently $d_n^3 b_n \in \mathbb{Z}$, holds [VERIFIED: $5 \leq p \leq 31$, $n \leq 320$, zero failures], with $v_p(d_n) = \lfloor \log_p n \rfloor$. But the corresponding equality $v_p(b_n/a_n) = -3v_p(d_n)$ is false: the numerators of b_n carry sporadic extra p -factors, for instance $7 \mid 62531 = \text{num}(b_3)$. The correct statement is an inequality.

3.4. The weight panel. Running the same experiment at lower weight explains both why the weight-three case is clean and where the danger lives. Let (A, B) be an Apéry-like pair with Apéry limit of weight w , and consider the master quantity $v_p(p^w B_n A_q - B_q A_n)$, $q = \lfloor n/p \rfloor$.

Observation 3.4 (congruence depth = weight). [VERIFIED: zero failures] the master floor is $\geq w$, flat, for all $p \geq 5$:

- $w = 1$: the central Delannoy numbers $1, 3, 13, 63, \dots$ with their companion, $B_n/A_n \rightarrow \log 2$; $n \leq 300$.
- $w = 2$: the Apéry pair for $\zeta(2)$, $(n+1)^2 u_{n+1} = (11n^2 + 11n + 3)u_n + n^2 u_{n-1}$, $A = 1, 3, 19, 147, 1251, \dots$; $n \leq 300$, floor exactly 2 for $p \geq 7$.
- $w = 3$: the pair (1.1)–(1.3), floor exactly 3 (Conjecture 3.1).

The panel also shows what the *ratio* form does at low weight, and it is instructive. At $w = 2$ the ratio descent $v_p(p^2 B_n/A_n - B_q/A_q)$ has a clean floor 2 at all residues except the reflection centre $r = (p-1)/2$, where the behaviour splits by p modulo 4: for $p \equiv 1 \pmod{4}$ it *boosts* (values 5, 8 at $p = 5$; 5 at $p = 13$), while for $p \equiv 3 \pmod{4}$ it *collapses to 0* ($p = 7, 11$) and the naive modulo- p statement fails outright. At $w = 1$ the floor is barely 1, the centre fails at $p = 7$ (depth 0) and $p = 11$ (depth -1), and there are sporadic dips to 0 elsewhere ($p = 13$ at $r = 2, 10$). Exceptional digits behave exactly as textbook: the case $p = 7$, $a = 3$ fails wholesale at $w = 2$ because $7^2 \mid A(3) = 147$; at $w = 1$ the failures sit at $(p, a) = (7, 3)$ with $7 \mid 63$, at $(13, 2)$ with $13 = A(2)$, and at $(3, 1)$ with $3 \mid 3$.

Observation 3.5 (the product form is the right universal statement). Every low-weight ratio-form failure just listed is a division artefact: it occurs exactly where p divides the Lucas factor $A(a)$. The product form

$$p^w B_{ap+r} \equiv B_a A_r \pmod{p}$$

holds at *every* previously failing site [VERIFIED: $w = 2$: $p = 7, 11$, all $a \leq 3$; $w = 1$: $p = 7, 11, 13$, all $a \leq 3$, including the cell $13 = A(2)$; $w = 3$: $p = 5$, $a \leq 3$].

Observation 3.5 is the reason this paper states its theorems in product form and not in ratio form, and it is what makes Theorem 2.18 valid at $p = 5$ (Remark 2.19). Historically, the programme first recorded a “dip to depth 2 at the reflection centre” at weight three; the full sweep showed that those cells are precisely the ratio-form pole cells — $11 \mid a_5 = 819005$, for example — and that on unit cells the depth floor is a flat 3. The correction is recorded here because it is the same phenomenon that governs the low-weight panel.

3.5. The general master form. Section 5 will show that at the level of all fifteen sporadic pairs one further correction is needed, a quadratic character χ . The general statement, of which Conjecture 3.1 is the case γ (with χ trivial and $w = 3$), is:

Conjecture 3.6 (master form (LB_w^χ)). *Let (A, B) be one of the fifteen sporadic Apéry-like pairs, w the weight of its Apéry limit and χ the quadratic character attached to it (Table 2). Then for every prime $p \geq 5$ and every $n \geq 1$, with $q = \lfloor n/p \rfloor$,*

$$v_p(p^w B_n A_q - \chi(p) B_q A_n) \geq w.$$

Evidence. [VERIFIED: zero failures] for all fifteen pairs, every prime $5 \leq p \leq 103$, all $n \leq 400$; and, for $p = 5, 7, 11$, all $n \leq 1200$ — five base- p digits at $p = 5$. Exact rational arithmetic throughout. The floor is exactly w in every single (sequence, prime) cell: perfectly flat, with no κ -dependence, no centre-residue exception and no exceptional-digit exception. The runs were organised as 15×16 cells for the primes $5 \leq p \leq 61$ with $n \leq 400$, plus 15×9 cells for the primes $67 \leq p \leq 103$ with $n \leq 400$, plus 15×3 cells for $p = 5, 7, 11$ with $n \leq 1200$. In the same sweeps the single-digit twisted Lucas form $p^w B_{ap+r} \equiv \chi(p) B_a A_r \pmod{p}$ and the integrality bound $v_p(B_n) \geq -w \lfloor \log_p n \rfloor$ each recorded zero failures.

The mod- p , single-digit shadow of Conjecture 3.6 is a theorem for a specified subclass; that is the content of Section 4.

4. THE GENERAL χ -TWISTED DESCENT

The proof of Theorem 2.18 in Section 2 is a two-page Kummer ledger. It turns out that the ledger is an artefact of Apéry's own choice of weight: if one replaces the nested alternating sum $\sum_{m \leq k} u(n, m)$ by a plain harmonic monomial — which Section 6 shows is possible, and in a startlingly short form — the whole argument collapses to three lines, and generalises at once to all weights, to several sequences, and to a twist by a quadratic character.

4.1. **Lemma K: Frobenius descent of a harmonic letter.**

Lemma 4.1 (Lemma K). [PROVED] *Let χ be a completely multiplicative arithmetic function (a Dirichlet character; $\chi \equiv 1$ is allowed), let $r \geq 1$ and $y \geq 0$, and set $K_\chi^{(r)}(y) = \sum_{m=1}^y \chi(m)m^{-r}$. Then, as an identity in $\mathbb{Q}$,*

$$p^r K_\chi^{(r)}(y) = \chi(p) K_\chi^{(r)}(\lfloor y/p \rfloor) + p^r T, \quad T := \sum_{\substack{m \leq y \\ p \nmid m}} \frac{\chi(m)}{m^r} \in \mathbb{Z}_{(p)}.$$

Proof. Split the sum according to $p \mid m$. The p -divisible part is

$$\sum_{j \leq \lfloor y/p \rfloor} \frac{\chi(jp)}{(jp)^r} = \chi(p) p^{-r} \sum_{j \leq \lfloor y/p \rfloor} \frac{\chi(j)}{j^r} = \chi(p) p^{-r} K_\chi^{(r)}(\lfloor y/p \rfloor),$$

using complete multiplicativity. The remaining terms have p -integral denominators, so their sum is $T \in \mathbb{Z}_{(p)}$. Multiply by p^r . $\square$

This single line is the entire source of the character in Conjecture 3.6. Sequences whose second solution needs χ -twisted harmonic letters acquire $\chi(p)$; sequences whose second solution is a pure ζ -harmonic monomial acquire $+1$.

Corollary 4.2 (Corollary K2). [PROVED] *Let $M = \prod_{t=1}^s K_{\chi_t}^{(r_t)}(x_t)$ with $\sum_t r_t = w$ and each $K_{\chi_t}^{(r_t)}(\lfloor x_t/p \rfloor) \in \mathbb{Z}_{(p)}$. Then*

$$p^w M \equiv \left(\prod_t \chi_t(p) \right) \prod_t K_{\chi_t}^{(r_t)}(\lfloor x_t/p \rfloor) \pmod{p}.$$

Proof. $p^w M = \prod_t (p^{r_t} K_{\chi_t}^{(r_t)}(x_t)) = \prod_t (\chi_t(p) K_{\chi_t}^{(r_t)}(\lfloor x_t/p \rfloor) + p^{r_t} T_t)$ by Lemma 4.1. Expanding, every term other than the pure product carries a factor p , and every factor is p -integral by hypothesis. $\square$

4.2. **Theorem LB.**

Definition 4.3 (admissible pair). Let $p \geq 5$. An *admissible pair* consists of

- a summand $S(n, k) = \prod_i \binom{L_i(n, k)}{M_i(n, k)}$ with L_i, M_i integer linear forms, and $A(n) = \sum_k S(n, k)$;
- a weight $w(n, k) = \sum_j c_j \prod_t K_{\chi}^{(r_{jt})}(x_{jt})$, a $\mathbb{Q}$ -combination of monomials of total degree $\sum_t r_{jt} = w$ in letters attached to one fixed quadratic character χ and to the trivial character, with the x_{jt} integer linear forms in (n, k) ;
- the second row $B(n) = \sum_k S(n, k)w(n, k)$.

Fix $1 \leq a < p$ and $0 \leq r, s < p$, and write $n = ap + r$, $k = bp + s$. The hypotheses are:

(H1): *Lucas/carry dichotomy.* For every k , either $p \mid S(n, k)$, or $S(n, k) \equiv S(a, b)S(r, s) \pmod{p}$.

(H2): *Product region.* The surviving set is $\{0 \leq b \leq a\} \times \Sigma_r$ for some $\Sigma_r \subseteq [0, p)$, and $\sum_{s \in \Sigma_r} S(r, s) \equiv A(r) \pmod{p}$.

- (H3): Digit compatibility.** On the surviving set, $\lfloor x(n, k)/p \rfloor = x(a, b)$ for every argument form x occurring in w .
- (H4): Tameness.** Every argument form satisfies $0 \leq x(n, k) \leq n$ on the summation range, and $c_j \in \mathbb{Z}_{(p)}$ for all j .
- (H5): χ -homogeneity.** Every monomial of w contains exactly the same number $e \in \{0, 1\}$ of χ -letters, all other letters carrying the trivial character.

Theorem 4.4 (Theorem LB). [PROVED] *Let $p \geq 5$ and let (S, w, A, B) be an admissible pair satisfying (H1)–(H5). Then for all $1 \leq a < p$ and $0 \leq r < p$, with $n = ap + r$,*

$$p^w B(ap + r) \equiv \chi(p)^e B(a) A(r) \pmod{p}.$$

Proof. By (H4), every argument satisfies $x(n, k) \leq n < p^2$, hence $\lfloor x/p \rfloor \leq a < p$ and every letter $K^{(r)}(\lfloor x/p \rfloor)$ is p -integral; likewise $x(a, b) \leq a < p$, so $w(a, b) \in \mathbb{Z}_{(p)}$. For $p^w w(n, k)$: each letter $K_\chi^{(r)}(x)$ with $x \leq n < p^2$ has $v_p \geq -r$ by Lemma 4.1, a monomial $\prod_t K_\chi^{(r_{jt})}$ of total degree w therefore has $v_p \geq -w$, and (H4) puts the coefficients c_j in $\mathbb{Z}_{(p)}$; hence $p^w w(n, k) \in \mathbb{Z}_{(p)}$ for every k . Corollary 4.2 then gives, for every k ,

$$p^w w(n, k) \equiv \chi(p)^e \tilde{w}(n, k) \pmod{p}, \quad \tilde{w}(n, k) := \sum_j c_j \prod_t K_\chi^{(r_{jt})}(\lfloor x_{jt}/p \rfloor), \quad (4.1)$$

and (H3) upgrades $\tilde{w}(n, k) = w(a, b)$ on the surviving set. Now split

$$p^w B(n) = \sum_{k: p|S(n, k)} S(n, k) p^w w(n, k) + \sum_{k \text{ surviving}} S(n, k) p^w w(n, k).$$

The first sum is $\equiv 0 \pmod{p}$, because each $p^w w(n, k)$ is p -integral and each $S(n, k)$ is divisible by p . In the second, substitute (H1) and (4.1):

$$p^w B(n) \equiv \chi(p)^e \sum_{b=0}^a \sum_{s \in \Sigma_r} S(a, b) S(r, s) w(a, b) = \chi(p)^e \left(\sum_{b=0}^a S(a, b) w(a, b) \right) \left(\sum_{s \in \Sigma_r} S(r, s) \right),$$

which by (H2) is $\equiv \chi(p)^e B(a) A(r) \pmod{p}$. $\square$

4.3. Remarks on the hypotheses.

Remark 4.5 (three lines instead of two pages). With $e = 0$, $S(n, k) = A(n, k)$ and the weight $w(n, k) = 2H_n^{(3)} - H_k^{(3)}$ of Theorem 6.1 below, Theorem 4.4 *reproves Theorem 2.18*. Hypotheses (H1)–(H3) for this instance are exactly Lemmas 2.5 and 2.10; (H4) holds because the arguments are n and k , both $\leq n$, and the coefficients are the integers $2, -1$; (H5) is vacuous. No Lemma 2.14, no Corollary 2.12, no Kummer ledger. The entire content of the long proof is absorbed into the choice of weight.

Remark 4.6 (what the mechanism actually needs). In the language of Section 2: “summand with Lucas factorisation and carry annihilation” is (H1)+(H2), which is available for all fifteen sporadic sequences because it is Malik–Straub’s theorem [13]; “harmonic weight of depth one” is the shape of w ; and the per-carry valuation bound $v_p \geq 2$ of Remark 2.15 is *not* needed — it is replaced by (H4), tameness, which makes $p^w w$ p -integral outright. Where tameness fails — because an argument form reaches $2n$, so that $p^w w(n, k)$ acquires a pole on the vanishing layer — one needs a substitute valuation bound; see Section 5.5.

Remark 4.7 (scope: single digit, modulo p). Theorem 4.4 gives the modulo- p , single-digit statement. The master form of Conjecture 3.6 — all n , modulo p^w — is [VERIFIED] but not proved here. The gap is exactly the one left open at weight three (Remark 2.21).

Corollary 4.8 (integrality). *Iterating the ratio form of Conjecture 3.6 over the base- p digits gives $v_p(B_n) \geq -w \lfloor \log_p n \rfloor$, i.e. $d_n^w B_n \in \mathbb{Z}_{(p)}$ — the coefficient-level shadow of p -integrality of the mirror map, see Section 8. Independent of the conjecture, this bound is [VERIFIED: 15 sequences $\times$ 16 primes $\times$ $n \leq 400$, zero failures].*

Remark 4.9 (the formalised hypotheses differ, and are weaker). The Lean 4 formalisation of Theorem 4.4 (Section 9) replaces (H1)+(H2) by the single unconditional congruence $S(ap + r, bp + s) \equiv S(a, b)S(r, s) \pmod{p}$ for all a, b together with the vanishing $S(n, k) = 0$ for $k > n$. In the carrying regime both sides vanish modulo p , so the dichotomy is absorbed and no product-region argument occurs anywhere in the machine proof; the double sum factors because a full block $\{0 \leq k < (a + 1)p\}$ splits, and the extension from $\{0 \leq k \leq n\}$ to the full block is an equality of integers rather than a congruence. It also replaces the counting form of (H5) by the equivalent but more flexible requirement that the product of the characters of the letters of each monomial, evaluated at p , be constant.

5. THE FIFTEEN SPORADIC FAMILIES

5.1. The pairs. Zagier’s search and its Almkvist–Zudilin and Cooper extensions produce fifteen sporadic Apéry-like sequences, satisfying one of

$$(R2) \quad (n + 1)^2 u_{n+1} = (an^2 + an + b)u_n - cn^2 u_{n-1}, \quad (5.1)$$

$$(R3) \quad (n + 1)^3 u_{n+1} = (2n + 1)(an^2 + an + b)u_n - n(cn^2 + d)u_{n-1}. \quad (5.2)$$

Zagier’s six satisfy (5.1); the six Almkvist–Zudilin sequences satisfy (5.2) with $d = 0$, and Cooper’s three satisfy (5.2) with $d \neq 0$. The sequence tables we use are those of Malik–Straub [13, Tables 1–2]; all fifteen binomial sums were [VERIFIED: $n \leq 11$] to reproduce the recurrence solution exactly.

The *second solution* $B(n)$ is normalised by $B(0) = 0$, $B(1) = 1$, with the recurrence enforced for $n \geq 1$ — the $n = 0$ step is degenerate. This is exactly Apéry’s construction, and the same normalisation used by Chamberland–Straub [7]. Rescaling B by a p -unit rational does not affect any statement below, so for $p \geq 5$ the normalisation is canonical up to harmless constants.

5.2. Limits, weights and integrality.

Observation 5.1 (Apéry limits). [VERIFIED] The limits in Table 2 were identified by two-term PSLQ at 400 digits against a 33-constant basis — $\zeta(2), \zeta(3), \zeta(4); L_D(2)$ and $L_D(3)$ for $D = -3, -4, -7, -8, -11, -15, -20, -24, 5, 8, 12, 13, 24; \log^2 2, \pi^2 \log 2, \log^3 2$ — matching to between 31 and 401 digits (the weakest is **F**, whose convergence rate is $(8/9)^n$; s_{18} converges at $(3/4)^n$). The known values are recovered: $\zeta(2)/4$ for Franel and $\zeta(2)/5$ for $\sum_k \binom{n}{k}^4$ (cf. [7, eq. (7.2)]) and $\zeta(3)/6$ for Apéry. The remaining eight — $7\zeta(3)/24$ (Domb), $7\zeta(3)/32$ (ε), $L_{-3}(3)/3$ (ζ), $\zeta(2)/7$ (s_7), $\frac{5}{8}L_{-3}(2)$ (**F**), $L_{-3}(2)/2$ (**C** and s_{18}) and $L_{-4}(2)/2$ (**E**, half Catalan’s constant) — are computed here. They are of the shape one expects (rational multiples of $\zeta(2)$, $L_{-3}(2)$, $L_{-4}(2)$), and some may well appear in existing numerical tables; we make no attribution claim.

Observation 5.2 (weights). [VERIFIED] Let w be the minimal exponent with $d_n^w B(n) \in \mathbb{Z}$, computed over $n \leq 600$. Then $w = 2$ for all six of Zagier’s (5.1) sequences and for all three of Cooper’s (5.2) sequences with $d \neq 0$, and $w = 3$ for all six Almkvist–Zudilin sequences ($d = 0$). It is sharp in every case: $d_n^{w-1} B_n$ is non-integral for at least 596 of the 600 values of n . In every case w equals the motivic weight of the Apéry limit. The local form $v_p(B_n) \geq -w \lfloor \log_p n \rfloor$ holds with zero failures over 15 sequences $\times$ 16 primes $\times$ $n \leq 400$.

TABLE 2. The fifteen sporadic pairs: parameters of (5.1)/(5.2), the integral solution $A(n)$, the Apéry limit $L = \lim B(n)/A(n)$, the weight w and the character χ .

#	label	$(a, b, c, d]$	$A(n)$	L	w	χ
1	A	$(7, 2, -8)$	$\sum_k \binom{n}{k}^3$ (Franel)	$\zeta(2)/4$	2	1
2	B	$(9, 3, 27)$	$\sum_k (-1)^k 3^{n-3k} \binom{n}{3k} \frac{(3k)!}{k!^3}$	none	2	χ_{-3}
3	C	$(10, 3, 9)$	$\sum_k \binom{n}{k}^2 \binom{2k}{k}$	$L_{-3}(2)/2$	2	χ_{-3}
4	D	$(11, 3, -1)$	$\sum_k \binom{n}{k}^2 \binom{n+k}{n-k}$	$\zeta(2)/5$	2	1
5	E	$(12, 4, 32)$	$\sum_k \binom{n}{k} \binom{2k}{k} \binom{2(n-k)}{n-k}$	$L_{-4}(2)/2$	2	χ_{-4}
6	F	$(17, 6, 72)$	$\sum_k (-1)^k 8^{n-k} \binom{n}{k} \sum_l \binom{k}{l}^3$	$\frac{5}{8} L_{-3}(2)$	2	χ_{-3}
7	(α)	$(10, 4, 64, 0)$	$\sum_k \binom{n}{k}^2 \binom{2k}{k} \binom{2(n-k)}{n-k}$ (Domb)	$7\zeta(3)/24$	3	1
8	(γ)	$(17, 5, 1, 0)$	$\sum_k \binom{n}{k}^2 \binom{n+k}{n}^2$ (Apéry)	$\zeta(3)/6$	3	1
9	(δ)	$(7, 3, 81, 0)$	$\sum_k (-1)^k 3^{n-3k} \binom{n}{3k} \binom{n+k}{n} \frac{(3k)!}{k!^3}$	none	3	1
10	(ε)	$(12, 4, 16, 0)$	$\sum_k \binom{n}{k}^2 \binom{2k}{n}^2$	$7\zeta(3)/32$	3	1
11	(ζ)	$(9, 3, -27, 0)$	$\sum_{k,l} \binom{n}{k}^2 \binom{n}{l} \binom{k}{l} \binom{k+l}{n}$	$L_{-3}(3)/3$	3	χ_{-3}
12	(η)	$(11, 5, 125, 0)$	$\sum_k (-1)^k \binom{n}{k}^3 \left[\binom{4n-5k-1}{3n} + \binom{4n-5k}{3n} \right]$	none	3	χ_5
13	s_7	$(13, 4, -27, 3)$	$\sum_k \binom{n}{k}^2 \binom{n+k}{k} \binom{2k}{n}$	$\zeta(2)/7$	2	1
14	s_{10}	$(6, 2, -64, 4)$	$\sum_k \binom{n}{k}^4$	$\zeta(2)/5$	2	1
15	s_{18}	$(14, 6, 192, -12)$	Cooper's s_{18}	$L_{-3}(2)/2$	2	χ_{-3}

Observation 5.3 (Cooper's three are lighter than their recurrence). s_7, s_{10}, s_{18} have weight 2, not 3, despite satisfying the cubic recurrence (5.2): the weight tracks the Cooper deformation $d \neq 0$, not the shape of the recurrence.

5.3. Where the untwisted law dies, and the repair.

Observation 5.4 (failure of the naive law). [VERIFIED] Testing $v_p(p^w B_n A_q - B_q A_n) \geq w$, $q = \lfloor n/p \rfloor$, over primes $5 \leq p \leq 43$ and $n \leq 300$ in exact arithmetic:

verdict	sequences
holds flat, every prime tested	A, D, $\alpha, \gamma, \delta, \varepsilon, s_7, s_{10}$ (8)
fails at $p \equiv 2 \pmod{3}$	B, C, F, ζ, s_{18} (5)
fails at $p \equiv 3 \pmod{4}$	E (1)
fails at $p \equiv \pm 2 \pmod{5}$	η (1)

The failures are total, not marginal: on the failing primes the single-digit floor is 0 — the difference is a p -unit — and in the multi-digit range the valuation drops to $-w(\#\text{digits} - 1)$, i.e. as low as the integrality bound permits.

The failing sets are not explained by exceptional digits ($p \mid A(a)$), nor by the centre residue, nor by $\kappa = v_p \binom{2n}{n}$: they are exactly the sets of primes at which a quadratic character takes the value -1 . Three cross-checks isolate the character [VERIFIED]: the passing primes of the χ_{-3} group are mixed modulo 4 and modulo 5; those of **E** are mixed modulo 3 and modulo 5; and those of η , namely 11, 19, 29, 31, 41, are mixed modulo 3 and modulo 4 and are precisely the primes $p \equiv \pm 1 \pmod{5}$.

Observation 5.5 (the repair is one sign). [VERIFIED: zero failures] At every cell with $\chi(p) = -1$, for all seven twisted sequences, all primes $5 \leq p \leq 31$ with $\chi(p) = -1$ and all $n < p^3$ (up to 1500),

$$v_p(p^w B_n A_q + B_q A_n) = w \quad \text{exactly.}$$

Observations 5.4 and 5.5 together are Conjecture 3.6. The character is the character of the Apéry limit — a correlation that is 12 out of 12 on the sequences whose limit exists — and Lemma 4.1 says why: the $\chi(p)$ in the congruence is the $\chi(p)$ that comes out of the p -divisible layer of a χ -twisted harmonic letter.

Observation 5.6 (ramified primes are benign). For η the prime $p = 5$ is ramified, $\chi_5(5) = 0$, and Conjecture 3.6 degenerates to $p^3 B_n A_q \equiv 0 \pmod{p^3}$ — which is exactly the assertion that B_n is 5-integral. [VERIFIED: $n \leq 500$] it is: $\min_n v_5(B_n) = 0$, so the pole simply does not develop at the ramified prime, and the floor is exactly 3. This is the only ramified prime ≥ 5 among the fifteen; the χ_{-3} and χ_{-4} sequences ramify only at 3 and 2.

5.4. Harmonic decompositions. To apply Theorem 4.4 one needs a harmonic-monomial weight. We found these by exact linear fitting: the ansatz is $B_n = \sum_k S(n, k)w(n, k)$ with S the sequence's own binomial summand and w a $\mathbb{Q}$ -combination of weight- w monomials in letters $H_x^{(r)}$ (and, when needed, $K_{\chi, x}^{(r)}$) at arguments x that are linear forms in (n, k) ; the systems were solved exactly modulo the prime $2^{61} - 1$, the coefficients rationally reconstructed, and then *validated exactly over $\mathbb{Q}$* on held-out values of n .

TABLE 3. Harmonic-monomial weights found (all depth one, all validated exactly over $\mathbb{Q}$ on held-out n). S is the summand from Table 2.

seq.	w	weight $w(n, k)$	held out
γ	3	$\frac{1}{3}H_n^{(3)} - \frac{1}{6}H_k^{(3)}$	$n = 60..72$
A	2	$\frac{1}{4}H_k^{(2)} + \frac{3}{4}H_k(H_k - H_{n-k})$	$n = 60..72$
D	2	$\frac{1}{5}[H_n^{(2)} + H_k(2H_k - H_{n-k} - H_n)]$	$n = 60..72$
s_{10}	2	$\frac{1}{5}H_k^{(2)} + \frac{4}{5}H_k(H_k - H_{n-k})$	$n = 60..72$
s_7	2	8 terms in $H^{(1,2)}$ at $\{k, n-k, n, 2k\}$	$n = 60..72$
ε	3	10 terms in $H^{(1,2,3)}$ at $\{k, n-k, 2k, 2k-n\}$	$n = 60..72$
α (Domb)	3	14 terms in $H^{(1,2,3)}$ at $\{k, n-k, 2k, 2n-2k\}$	$n = 60..72$
E	2	$\frac{1}{2}K_{2k}^{(2)} + \frac{3}{4}H_k(K_{2k} - K_{2n-2k}) - \frac{1}{2}H_{2k}(K_{2k} - K_{2n-2k})$, $K = K_{\chi_{-4}}$	$n = 70..82$

The γ line, multiplied by 6, is the minimal closed form (1.4); Section 6 proves it. Two structural comments.

Remark 5.7 (a uniform shape for $\sum_k \binom{n}{k}^d$). **A** and s_{10} have literally the same shape, $\frac{1}{d+1}H_k^{(2)} + \frac{d}{d+1}H_k(H_k - H_{n-k})$ for $A^{(d)}(n) = \sum_k \binom{n}{k}^d$ with $d = 3, 4$. This is the natural source of Chamberland–Straub’s conjecture [7, eq. (7.4)] that the Apéry limit of $A^{(d)}$ is $\zeta(2)/(d+1)$, and suggests it for every d [VERIFIED: $d = 3, 4$ only; the recurrences for $d \geq 5$ have order ≥ 3 and were not tested].

Observation 5.8 (negative results, and the dichotomy). [VERIFIED] Exact inconsistency of the fitting systems:

- **C**: no decomposition on the summand $\binom{n}{k}^2 \binom{2k}{k}$ exists in *any* basis tried — plain harmonic monomials at $\{n, k, n-k, 2k\}$ (14 elements), at $\{n, k, n-k, 2k, n+k, 2n-2k, 2n, 2n-k\}$ (44 elements), and with χ_{-3} -letters adjoined (44 and 152 elements, 110 and 220 exact equations). All inconsistent.
- **E**: inconsistent in every *pure* harmonic basis tried (up to 44 elements, 8 arguments), and consistent the moment χ_{-4} -letters are allowed. This is the decisive experiment.
- δ : inconsistent in the weight-3 basis at $\{n, k, 3k, n-3k, n+k\}$ (65 elements, 110 equations).

- **B, F, ζ , η , s_{18}** : not attempted (double or alternating sums; ζ needs a genuine two-index ansatz).

The dichotomy behind this is elementary but decisive. A pure harmonic monomial $\prod_t H_{x_t}^{(r_t)}$ can only ever produce values in the ring generated by $\zeta(2), \zeta(3), \dots$ in the limit; it can never produce $L(\chi, w)$. Hence *no* χ -twisted sequence can have a pure- ζ harmonic decomposition, and the letters must be twisted — confirmed by **E**, and the reason **C**’s plain fits are inconsistent. **C**’s residual obstruction is that its conductor-3 letters need arguments that $\binom{n}{k}^2 \binom{2k}{k}$ does not provide; the right move there is a constant-term representation, not a bigger monomial basis [11]. This remains [OPEN].

5.5. Verdict: which of the fifteen are proved. We now state precisely which families Theorem 4.4 covers. *The labels below are the paper’s evidence classes and are not to be read as claims of uniform coverage.*

[PROVED] — hypotheses (H1)–(H5) verified, Theorem 4.4 applies.

seq.	χ	primes	why (H4) holds
γ (Apéry $\zeta(3)$)	1	$p \geq 5$	arguments $\{n, k\}$; coefficients $\frac{1}{3}, \frac{1}{6}$
A (Franel)	1	$p \geq 5$	arguments $\{k, n - k\}$; coefficients $\frac{1}{4}, \frac{3}{4}$
D (Apéry $\zeta(2)$) [†]	1	$p \geq 7$	arguments $\{n, k, n - k\}$; coefficients $\frac{1}{5}, \frac{2}{5}$
s_{10} $(\sum_k \binom{n}{k}^4)$	1	$p \geq 7$	arguments $\{k, n - k\}$; coefficients $\frac{1}{5}, \frac{4}{5}$

[†]One qualification, and it applies to **D** alone. What Theorem 4.4 controls is the explicit harmonic sum B . For γ , **A** and s_{10} the identity $(LB) = 0$ — which says that this B is the sequence’s second solution — is [PROVED] in Section 6. For **D** it is not: it is [VERIFIED: $n \leq 25$] only, and Proposition 6.23 shows that the template used for the other three provably cannot supply it. So the **D** row is [PROVED] as a statement about the explicit sum B , and [VERIFIED] as a statement about the $\zeta(2)$ -Apéry second solution.

For these, (H1)–(H3) hold by the argument of Section 2: for $S = \binom{n}{k}^\alpha \binom{n+k}{n}^\beta$ the surviving set is $\{s \leq r\}$ (no borrow in $\binom{n}{k}$) intersected with $\{r + s < p\}$ (no carry in $\binom{n+k}{n}$), a product region; there $\binom{n}{k} \equiv \binom{a}{b} \binom{r}{s}$ and $\binom{n+k}{n} \equiv \binom{a+b}{a} \binom{r+s}{r}$ by Lemma 2.3, and the deleted $r + s \geq p$ terms of $A(r)$ vanish modulo p — this is verbatim Lemma 2.10. Digit compatibility holds because $\lfloor k/p \rfloor = b$, $\lfloor (n - k)/p \rfloor = a - b$ (no borrow) and $\lfloor n/p \rfloor = a$.

The two steps of the proof of Theorem 4.4 — “the vanishing layer is $\equiv 0$ ” and “the surviving layer is $\equiv \chi(p)^e B_a A_r$ ” — were also checked separately and exactly on all cells $n = ap + r < p^2$: zero failures for γ , **A**, s_{10} at $p = 5, 7, 11, 13$ (both layers); zero failures for **D** at $p = 7, 11, 13$, but 11 failing cells at $p = 5$, which is exactly the (H4) coefficient condition biting (**D**’s coefficients are $\frac{1}{5}(1, 2, -1, -1)$). So **D**’s proof is unconditional for $p \geq 7$, and $p = 5$ is [VERIFIED] only. s_{10} ’s coefficients are also $\frac{1}{5}(1, 4, -4)$ yet it exhibits no defect at $p = 5$ — the layer valuations happen to be one higher there — but the proof as written still needs $p \geq 7$ for s_{10} . No other defect of any kind appears in the tame cases: the only obstruction the mechanism ever meets is the (H4) coefficient condition.

Need the Lemma-D upgrade — decomposition known, tameness fails. For α (Domb), ε , s_7 and **E** a decomposition is known (Table 3), but (H4) fails: the arguments reach $2n$ (letters at $2k, 2n - 2k, 2k - n$), so $p^w w(n, k)$ acquires a pole on the vanishing layer and the two-layer split of the proof of Theorem 4.4 is *invalid termwise*. [VERIFIED]: the vanishing-layer and surviving-layer defects occur in exactly equal numbers and cancel, which is why the congruence is nonetheless true. What is needed is a valuation bound showing that a Kummer carry in the wide binomial $(\binom{2k}{k}, \binom{2k}{n})$ beats the order-one pole of $H^{(r)}(\lfloor 2k/p \rfloor)$. (s_7 additionally excludes

$p = 7$: its coefficients have denominator 28; [VERIFIED] 13 failing cells at $p = 7$ and none at $p = 5, 11, 13$.) Notably **E** is the only known instance of the twisted theorem ($e = 1$), and it is non-tame: attempts to find a *tame* χ_{-4} -decomposition for **E** (arguments $\{n, k, n - k\}$, and $\{n, k, n - k, 2k\}$) were [VERIFIED: inconsistent] — the wide arguments are essential. So the twisted case of Theorem 4.4 is proved as a mechanism but has, at present, no tame instance. This is [OPEN].

Remain conjectural. For **B**, **C**, **F**, δ , ζ , η and s_{18} no harmonic decomposition is known, and Conjecture 3.6 is [VERIFIED] only. Of these, **C** and δ carry *proved-negative* results in the natural bases (Observation 5.8), so they need genuinely new letters: for **C** the conductor-3 letters, for δ presumably letters attached to the 3-section of the summand.

5.6. The sequences with no archimedean Apéry limit.

Observation 5.9. [VERIFIED] Three of the fifteen have no Apéry limit at all. For (5.1) the characteristic roots solve $\lambda^2 - a\lambda + c = 0$, and for (5.2), $\lambda^2 - 2a\lambda + c = 0$. The discriminants are

$$\mathbf{B} : -27, \quad \delta : -128, \quad \eta : -16,$$

i.e. complex-conjugate pairs of equal modulus, so Poincaré–Perron provides no dominant solution and $B(n)/A(n)$ does not converge — numerically, 0 digits of agreement between $n = 590$ and $n = 600$, against 380 digits for the convergent sequences. Every other sporadic sequence has real distinct roots.

Yet all three satisfy Conjecture 3.6 with a flat floor, with $\chi = \chi_{-3}$, trivial, and χ_5 respectively. *The p -adic Apéry limit exists where the archimedean one does not.* This is the cleanest evidence available to us that the descent is a statement about Frobenius and not about the archimedean limit; we return to it in Section 8.

6. MINIMAL CLOSED FORMS FOR SECOND SOLUTIONS

Theorem 4.4 needs a harmonic-monomial weight. For Apéry’s b_n the fitting procedure of Section 5.4 returned one of startling brevity, and this section proves that it is correct. We then prove two more closed forms of the same kind, and record a sharp obstruction: the method provably fails for the $\zeta(2)$ -Apéry sequence.

6.1. The minimal form.

Theorem 6.1 (minimal form). [PROVED] *For every integer $n \geq 0$,*

$$\sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 (2H_n^{(3)} - H_k^{(3)}) = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 \left(H_n^{(3)} + \sum_{m=1}^k \frac{(-1)^{m-1}}{2m^3 \binom{n}{m} \binom{n+m}{m}} \right) = b_n.$$

Equivalently, $\sum_{k=0}^n A(n, k)(H_n^{(3)} - H_k^{(3)} - \sum_{m=1}^k u(n, m)) = 0$.

Write $B_{\min}(n)$ for the left-hand sum. The proof is by the classical route: both sides satisfy Apéry’s recurrence and agree at $n = 0, 1$. Set

$$P(n) := 34n^3 + 51n^2 + 27n + 5, \quad (Lu)(n) := (n+1)^3 u(n+1) - P(n)u(n) + n^3 u(n-1), \quad (6.1)$$

and, for integers $n \geq 1$, $k \geq 0$, the *base term*

$$\Phi(n, k) := \frac{\binom{n+1}{k}^2 \binom{n+k-1}{k}^2}{n^2(n+1)^2}. \quad (6.2)$$

Lemma 6.2 (conversion). [PROVED] *For all integers $n \geq 1$, $k \geq 0$:*

$$A(n, k) = \Phi(n, k)(n+1-k)^2(n+k)^2, \quad (0a)$$

$$A(n+1, k) = \Phi(n, k)(n+k)^2(n+k+1)^2, \quad (0b)$$

$$A(n-1, k) = \Phi(n, k)(n+1-k)^2(n-k)^2, \quad (0c)$$

$$\Phi(n, k+1)(k+1)^4 = \Phi(n, k)(n+1-k)^2(n+k)^2. \quad (0d)$$

Proof. Each reduces to the binomial conversions $\binom{n}{k}(n+1) = \binom{n+1}{k}(n+1-k)$, $\binom{n+k}{k}n = \binom{n+k-1}{k}(n+k)$, $\binom{n+k+1}{k}n(n+1) = \binom{n+k-1}{k}(n+k)(n+k+1)$, $\binom{n-1}{k}n(n+1) = \binom{n+1}{k}(n+1-k)(n-k)$, and, for (0d), $\binom{n+1}{k+1}(k+1) = \binom{n+1}{k}(n+1-k)$, $\binom{n+k}{k+1}(k+1) = \binom{n+k-1}{k}(n+k)$. The last two, those of (0d), are absorption identities $\binom{N}{j+1}(j+1) = \binom{N}{j}(N-j)$ — the second of them followed by one step of the companion. The first four move the *upper* index instead and are instances of that companion, $\binom{N+1}{j}(N+1-j) = \binom{N}{j}(N+1)$, applied once or twice. All six hold for *every* integer $k \geq 0$, including the degenerate range $k > n$ where both sides vanish: the conversions are stated multiplicatively, so no 0/0 arises. $\square$

Lemma 6.2 is the device that removes every boundary subtlety: after it, all identities below are $\Phi(n, k)$ times polynomial identities in $\mathbb{Q}[n, k]$. [VERIFIED: all six conversions, $n = 1..12$, $k = 0..n+4$, difference exactly 0].

6.2. Step B: the Zeilberger certificate for a_n . Set

$$T(n, k) := 4(2n+1)(2k^2 - 3k - 4n^2 - 4n), \quad G(n, k) := \Phi(n, k)k^4T(n, k). \quad (6.3)$$

Lemma 6.3 (CERT-1). [PROVED] *As an identity in $\mathbb{Q}[n, k]$,*

$$\begin{aligned} & T(n, k+1)(n+1-k)^2(n+k)^2 - k^4T(n, k) \\ &= (n+1)^3(n+k+1)^2(n+k)^2 - P(n)(n+1-k)^2(n+k)^2 + n^3(n-k)^2(n+1-k)^2. \end{aligned}$$

Proposition 6.4. [PROVED] *For all integers $n \geq 1$, $k \geq 0$,*

$$(n+1)^3A(n+1, k) - P(n)A(n, k) + n^3A(n-1, k) = G(n, k+1) - G(n, k). \quad (6.4)$$

Proof. By (0a)–(0c) the left side is $\Phi(n, k)$ times the right-hand side of Lemma 6.3. By (0d), $G(n, k+1) = \Phi(n, k+1)(k+1)^4T(n, k+1) = \Phi(n, k)(n+1-k)^2(n+k)^2T(n, k+1)$ and $G(n, k) = \Phi(n, k)k^4T(n, k)$, so the right side is $\Phi(n, k)$ times the left-hand side of Lemma 6.3. $\square$

The boundary values are $G(n, 0) = 0$ (factor k^4), $G(n, k) = 0$ for $k \geq n+2$ (factor $\binom{n+1}{k}^2$), and $G(n, n+1) = -4(n+1)^3\binom{2n+1}{n+1}^2$. Summing (6.4) over $k = 0, \dots, N$ with $N \geq n+1$ telescopes to $G(n, N+1) - G(n, 0) = 0$, which recovers Apéry's recurrence $(La)(n) = 0$.

6.3. Step B' and Step C: L annihilates $B_{\min}$. Write $w(n, k) := 2H_n^{(3)} - H_k^{(3)}$, so $B_{\min}(n) = \sum_{k \geq 0} A(n, k)w(n, k)$ (terms with $k > n$ vanish). Since $w(n+1, k) = w(n, k) + 2/(n+1)^3$ and $w(n-1, k) = w(n, k) - 2/n^3$,

$$(LB_{\min})(n) = \sum_{k \geq 0} w(n, k)(G(n, k+1) - G(n, k)) + 2a_{n+1} - 2a_{n-1},$$

by Proposition 6.4. Abel summation, with $w(n, k+1) - w(n, k) = -1/(k+1)^3$ and the boundary values of G , turns this into

$$(LB_{\min})(n) = \sum_{j \geq 1} \frac{G(n, j)}{j^3} + 2a_{n+1} - 2a_{n-1}. \quad (6.5)$$

Now set

$$U(n, k) := -4(k-1)(2n+1)(2n^2+2n+1-k), \quad K(n, k) := \frac{\Phi(n, k)k^4 U(n, k)}{n^2(n+1)^2}. \quad (6.6)$$

Lemma 6.5 (CERT-2). [PROVED] *As an identity in $\mathbb{Q}[n, k]$,*

$$\begin{aligned} & U(n, k+1)(n+1-k)^2(n+k)^2 - k^4 U(n, k) \\ &= n^2(n+1)^2 \left[kT(n, k) + 2((n+k)^2(n+k+1)^2 - (n+1-k)^2(n-k)^2) \right], \end{aligned}$$

and the bracket equals $4k(2n+1)(4k^2-3k-2n-2n^2)$.

Proposition 6.6. [PROVED] *For all integers $n \geq 1$, $k \geq 0$, with the convention $G(n, 0)/0^3 := 0$ (consistent, since $G(n, k)/k^3 = \Phi(n, k)kT(n, k)$),*

$$\frac{G(n, k)}{k^3} + 2A(n+1, k) - 2A(n-1, k) = K(n, k+1) - K(n, k). \quad (6.7)$$

Consequently $\sum_{j \geq 1} G(n, j)/j^3 + 2a_{n+1} - 2a_{n-1} = 0$, since $K(n, 0) = 0$ and $K(n, k) = 0$ for $k \geq n+2$.

Proof. By (0b), (0c) the left side is $\Phi(n, k)[kT(n, k) + 2((n+k)^2(n+k+1)^2 - (n+1-k)^2(n-k)^2)]$; by (0d) the right side is $\Phi(n, k)[(n+1-k)^2(n+k)^2 U(n, k+1) - k^4 U(n, k)]/(n^2(n+1)^2)$. Equality is Lemma 6.5. Summing over $k = 0, \dots, N$ with $N \geq n+1$ telescopes. $\square$

Theorem 6.7. [PROVED] $(LB_{\min})(n) = 0$ for all $n \geq 1$.

Proof. Combine (6.5) with Proposition 6.6. $\square$

6.4. Step D: the classical companion satisfies the same recurrence. This is the half in which Apéry's harmonic weight $c(n, k)$ must be differenced *in* n . Everything collapses because of two one-line Gosper certificates whose entire content is the pair of polynomial identities

$$m^2 + (n+m)(n-m) = n^2, \quad m^2 + (n+1-m)(n+1+m) = (n+1)^2. \quad (6.8)$$

Put $D(n, k) := \binom{n}{k} \binom{n+k}{k}$ (so $A = D^2$), and

$$\begin{aligned} \varphi(n, m) &:= m u(n, m) = \frac{(-1)^{m-1}((m-1)!)^2(n-m)!}{2(n+m)!}, \\ \Lambda(n, k) &:= (n-k)\varphi(n, k+1) = \frac{(-1)^k}{2(n+k+1)D(n, k)}. \end{aligned}$$

Since $D(n, k) \neq 0$ for $0 \leq k \leq n$, every division below is legitimate on its stated range.

Lemma 6.8 (the two inner Gosper certificates). [PROVED] *For $1 \leq m \leq n$ put*

$$Y_1(m) := -\frac{(n+m)\varphi(n, m)}{n^2}, \quad Y_2(m) := -\frac{(n+1-m)\varphi(n, m)}{(n+1)^2}.$$

Then, using $\varphi(n, m+1)/\varphi(n, m) = -m^2/((n-m)(n+m+1))$,

$$Y_1(m+1) - Y_1(m) = \frac{\varphi(n, m)}{n-m} \quad (1 \leq m \leq n-1), \quad Y_2(m+1) - Y_2(m) = \frac{\varphi(n, m)}{n+m+1} \quad (1 \leq m \leq n),$$

the two brackets collapsing by the identities (6.8).

Lemma 6.9 (the n -differences of Apéry's weight). [PROVED] Using $u(n+1, m) = u(n, m) \frac{n+1-m}{n+m+1}$ and $u(n-1, m) = u(n, m) \frac{n+m}{n-m}$, i.e. $u(n+1, m) - u(n, m) = -2\varphi(n, m)/(n+m+1)$ and $u(n-1, m) - u(n, m) = 2\varphi(n, m)/(n-m)$, Lemma 6.8 telescopes with $Y_2(1) = -\frac{1}{2(n+1)^3}$, $Y_1(1) = -\frac{1}{2n^3}$ to give

$$c(n+1, k) - c(n, k) = \frac{2\Lambda(n, k)}{(n+1)^2} \quad (0 \leq k \leq n),$$

$$c(n-1, k) - c(n, k) = -\frac{2(n+k+1)\Lambda(n, k)}{(n-k)n^2} \quad (0 \leq k \leq n-1).$$

Remark 6.10 (Apéry's miracle). In Lemma 6.9 the rational pieces $1/(n+1)^3$ and $-1/n^3$ coming from $H_{n\pm 1}^{(3)} - H_n^{(3)}$ cancel *exactly* against the boundary values $Y_2(1)$, $Y_1(1)$ of the inner Gosper antidifferences, and that cancellation is precisely the polynomial identity $m^2 + (n+1-m)(n+m+1) = (n+1)^2$. This is why Apéry's weight carries the $\frac{1}{2}$, the $(-1)^{m-1}$ and the $m^3 \binom{n}{m} \binom{n+m}{m}$: the weight is not found, it is *forced* — $c(n, k)$ is the unique weight, up to the kernel of the recurrence, whose n -difference is a *single hypergeometric term* rather than a sum.

Lemma 6.11 (the correction term). [PROVED] For $0 \leq k \leq n$ set $\mathcal{E}(n, k) := (n+1)^3 A(n+1, k)(c(n+1, k) - c(n, k)) + n^3 A(n-1, k)(c(n-1, k) - c(n, k))$, the second summand being 0 at $k = n$ because $A(n-1, n) = 0$. Then by Lemma 6.9, (0b), (0c) and $A(n, k)\Lambda(n, k) = (-1)^k D(n, k)/(2(n+k+1))$,

$$\mathcal{E}(n, k) = (-1)^k D(n, k) \left[\frac{(n+1)(n+k+1)}{(n+1-k)^2} - \frac{n(n-k)}{(n+k)^2} \right].$$

Proposition 6.12. [PROVED] For $0 \leq k \leq n$, with $B(n-1, k) := 0$ for $k \geq n$ and $B(n, k) := A(n, k)c(n, k)$,

$$(n+1)^3 B(n+1, k) - P(n)B(n, k) + n^3 B(n-1, k) = c(n, k)(G(n, k+1) - G(n, k)) + \mathcal{E}(n, k).$$

At $k = n+1$ only the first term survives: the left side equals $(n+1)^3 A(n+1, n+1)c(n+1, n+1)$.

Proof. Substitute $c(n \pm 1, k) = c(n, k) + (c(n \pm 1, k) - c(n, k))$ and use Proposition 6.4. $\square$

Lemma 6.13 (the top term). [PROVED] $G(n, n+1) = -4(n+1)^3 \binom{2n+1}{n+1}^2 = -(n+1)^3 A(n+1, n+1)$, using $\binom{2n+2}{n+1} = 2\binom{2n+1}{n+1}$. Moreover $u(n+1, n+1) = \frac{(-1)^n}{4(n+1)^3 \binom{2n+1}{n+1}}$ and $\Lambda(n, n) = \frac{(-1)^n}{2(n+1) \binom{2n+1}{n+1}}$, so that $c(n+1, n+1) - c(n, n) = \frac{5}{4} (-1)^n / ((n+1)^3 \binom{2n+1}{n+1})$ and

$$\text{Top}(n) := c(n, n)G(n, n+1) + (n+1)^3 A(n+1, n+1)c(n+1, n+1) = 5(-1)^n \binom{2n+1}{n+1}.$$

This is where the classical 5 of $\zeta(3) = \frac{5}{2} \sum_m (-1)^{m-1} m^{-3} \binom{2m}{m}^{-1}$ enters.

Summing Proposition 6.12 over $k = 0, \dots, n$, adding the $k = n+1$ term, and Abel-summing $\sum_{k=0}^n c(n, k)(G(n, k+1) - G(n, k)) = c(n, n)G(n, n+1) - \sum_{j=1}^n u(n, j)G(n, j)$ (using $G(n, 0) = 0$), we get

$$(Lb)(n) = \text{Top}(n) + \sum_{k=0}^n V(n, k), \quad V(n, k) := \mathcal{E}(n, k) - u(n, k)G(n, k), \quad (6.9)$$

the $k = 0$ term of the second piece being 0. Since

$$u(n, k)G(n, k) = \frac{(-1)^{k-1} D(n, k) k T(n, k)}{2(n+1-k)^2 (n+k)^2},$$

we have $V(n, k) = (-1)^k D(n, k)W(n, k)$ with

$$W(n, k) = \frac{(n+1)(n+k+1)}{(n+1-k)^2} - \frac{n(n-k)}{(n+k)^2} + \frac{kT(n, k)}{2(n+1-k)^2(n+k)^2}.$$

Lemma 6.14 (CERT-3). [PROVED] *As an identity in $\mathbb{Q}[n, k]$,*

$2(n+1)(n+k+1)(n+k)^2 - 2n(n-k)(n+1-k)^2 + kT(n, k) + 10k(2n+1)(n+1-k)(n+k) = 0$,
that is,

$$W(n, k) = \frac{-5k(2n+1)}{(n+1-k)(n+k)}, \quad \text{whence}$$

$$V(n, k) = \frac{-5(2n+1)}{n(n+1)} (-1)^k k \binom{n+1}{k} \binom{n+k-1}{k},$$

using $\binom{n}{k}/(n+1-k) = \binom{n+1}{k}/(n+1)$ and $\binom{n+k}{k}/(n+k) = \binom{n+k-1}{k}/n$.

The entire weight-three structure thus collapses to a single linear-over-quadratic rational function.

Lemma 6.15 (residual binomial identity). [PROVED]

$$(D\text{-}BIN) \quad \sum_{k=0}^n (-1)^k k \binom{n+1}{k} \binom{n+k-1}{k} = (-1)^n n \binom{2n}{n}.$$

Proof. Put $s(n, k) := (-1)^k k \binom{n+1}{k} \binom{n+k-1}{k}$ and $Y(n, k) := -k(k-1)s(n, k)/(n(n+1))$. The absorption identities $\binom{n+1}{k+1}(k+1) = \binom{n+1}{k}(n+1-k)$ and $\binom{n+k}{k+1}(k+1) = \binom{n+k-1}{k}(n+k)$ of Lemma 6.2 give $(k+1)s(n, k+1) = -(-1)^k \binom{n+1}{k} \binom{n+k-1}{k} (n+1-k)(n+k)$, hence for every integer $k \geq 0$, division-free,

$$\begin{aligned} n(n+1)(Y(n, k+1) - Y(n, k)) \\ = (-1)^k k \binom{n+1}{k} \binom{n+k-1}{k} [(n+1-k)(n+k) + k(k-1)] = n(n+1)s(n, k), \end{aligned}$$

by the polynomial identity $(n+1-k)(n+k) + k(k-1) = n(n+1)$. So $Y(n, \cdot)$ telescopes $s(n, \cdot)$ and $\sum_{k=0}^n s(n, k) = Y(n, n+1) - Y(n, 0) = Y(n, n+1) = -s(n, n+1) = (-1)^n n \binom{2n}{n}$, using $\binom{2n}{n+1} = n \binom{2n}{n}/(n+1)$. $\square$

Theorem 6.16. [PROVED] $(Lb)(n) = 0$ for all $n \geq 1$.

Proof. By (6.9), Lemma 6.14 and Lemma 6.15,

$$\begin{aligned} \sum_{k=0}^n V(n, k) &= \frac{-5(2n+1)}{n(n+1)} (-1)^n n \binom{2n}{n} = -5(-1)^n \binom{2n}{n} \frac{2n+1}{n+1} \\ &= -5(-1)^n \binom{2n+1}{n+1} = -\text{Top}(n). \end{aligned} \quad \square$$

Proof of Theorem 6.1. Theorems 6.7 and 6.16 say that both sides are annihilated by L , whose leading coefficient $(n+1)^3$ never vanishes for $n \geq 1$. It remains to check agreement at $n = 0, 1$. We have $B_{\min}(0) = A(0, 0)(2H_0^{(3)} - H_0^{(3)}) = 0 = b_0$, and $B_{\min}(1) = A(1, 0) \cdot 2 + A(1, 1) \cdot 1 = 1 \cdot 2 + 4 \cdot 1 = 6$; on the other side $u(1, 1) = 1/(2 \cdot 1^3 \binom{1}{1} \binom{2}{1}) = 1/4$, so $c(1, 0) = 1$, $c(1, 1) = 5/4$ and $b_1 = 1 \cdot 1 + 4 \cdot \frac{5}{4} = 6$. Induct. $\square$

TABLE 4. Certificate inventory for Theorem 6.1. All are verified by exact polynomial arithmetic (**Expand**→ 0).

#	role	certificate
CERT-1	Zeilberger certificate for a_n	$T(n, k) = 4(2n + 1)(2k^2 - 3k - 4n^2 - 4n)$, $G = \Phi k^4 T$
CERT-2	Gosper certificate for the w -correction	$U(n, k) = -4(k - 1)(2n + 1)(2n^2 + 2n + 1 - k)$, $K = \Phi k^4 U / (n^2(n + 1)^2)$
CERT-3	the W -collapse	$W(n, k) = -5k(2n + 1) / ((n + 1 - k)(n + k))$
CERT-4	inner Gosper certificate, Δ_n^+	$Y_2(m) = -(n + 1 - m)\varphi(n, m) / (n + 1)^2$
CERT-5	inner Gosper certificate, Δ_n^-	$Y_1(m) = -(n + m)\varphi(n, m) / n^2$
CERT-6	Gosper certificate for (D-BIN)	$Y(n, k) = -k(k - 1)s(n, k) / (n(n + 1))$

The supporting polynomial identities, all **Expand**→ 0, are $m^2 + (n + m)(n - m) = n^2$; $m^2 + (n + 1 - m)(n + m + 1) = (n + 1)^2$; $(n + 1 - k)(n + k) + k(k - 1) = n(n + 1)$; and CERT-1, CERT-2, CERT-3 as displayed. Numerically, $B_{\min}(n) - b_n = 0$ exactly for $n = 0..50$ in exact rational arithmetic, in two independent implementations [VERIFIED: $n \leq 50$]; and $B_{\min}$ reproduces the literature values $0, 6, \frac{351}{4}, \frac{62531}{36}, \frac{11424695}{288}, \frac{35441662103}{36000}$.

Remark 6.17 (formalisation-friendliness). Nothing above uses Gamma functions, limits or $0/0$: Lemma 6.2 converts every binomial ratio into a *multiplicative* absorption identity valid for all integers $k \geq 0$, after which each step is a polynomial identity in $\mathbb{Q}[n, k]$ plus a finite telescoping sum. The only analytic input is $H_{n+1}^{(3)} - H_n^{(3)} = 1/(n + 1)^3$.

6.5. Two more closed forms, and the mechanism behind them.

Observation 6.18 (symmetrisation). [PROVED] For the d -fold binomial sums $A^{(d)}(n) = \sum_k \binom{n}{k}^d$ the summand is symmetric under $k \mapsto n - k$, so the fitted weights of Table 3 may be replaced by their $k \leftrightarrow n - k$ averages without changing the sum:

$$\frac{1}{d+1} H_k^{(2)} + \frac{d}{d+1} H_k(H_k - H_{n-k}) \rightsquigarrow \frac{1}{2(d+1)} (H_k^{(2)} + H_{n-k}^{(2)}) + \frac{d}{2(d+1)} (H_k - H_{n-k})^2.$$

[VERIFIED: exactly, $n = 0..12$, $d = 3, 4$]. Concretely,

$$B_{\mathbf{A}}(n) = \sum_k \binom{n}{k}^3 \left[\frac{1}{8} (H_k^{(2)} + H_{n-k}^{(2)}) + \frac{3}{8} (H_k - H_{n-k})^2 \right],$$

$$B_{s_{10}}(n) = \sum_k \binom{n}{k}^4 \left[\frac{1}{10} (H_k^{(2)} + H_{n-k}^{(2)}) + \frac{2}{5} (H_k - H_{n-k})^2 \right].$$

This is not cosmetic. In the symmetric form the only weight-one letter that ever appears is the single antisymmetric combination $\delta(n, k) := H_k - H_{n-k}$, because $\sigma := H_k^{(2)} + H_{n-k}^{(2)}$ differences to *rational* functions under both $n \mapsto n \pm 1$ and $k \mapsto k + 1$. The certificate module is therefore of rank 2 ($\{1, \delta\}$) instead of rank 4 ($\{1, H_k, H_{n-k}, H_n\}$), and one faces two Gosper problems instead of four.

The template is then: (i) a Zeilberger certificate $G(n, k) = S(n, k)k^d \rho(n, k) / (n + 1 - k)^d$, whose defining equation collapses to one polynomial equation in $\mathbb{Q}[n, k]$ because $S(n, k + 1) / S(n, k) = ((n - k) / (k + 1))^d$, so that the shifted certificate is the summand times a polynomial; (ii) a split of the recurrence applied to $S \cdot w$ into $w \cdot \Delta_k G$ plus a correction whose n -differences lie in

$\mathbb{Q}(n, k) + \mathbb{Q}(n, k)\delta$; (iii) Abel summation; (iv) two Gosper problems

$$r(n, k)z_1(n, k+1) - z_1(n, k) = B(n, k),$$

$$r(n, k)z_0(n, k+1) - z_0(n, k) = A(n, k) - r(n, k)z_1(n, k+1)e(n, k),$$

with $r = S(n, k+1)/S(n, k)$ and $e = \frac{1}{k+1} + \frac{1}{n-k}$, both solved by rational ansätze $z_1 = N(k)/(n+1-k)^p$, $z_0 = M(k)/(n+1-k)^{p+1}$; and (v) three scalar boundary identities, of which the first,

$$(b1) \quad \rho(n, n+1) = -c_+(n) S(n+1, n+1)/S(n, n),$$

is structural: it says that the Zeilberger certificate at the top of its support reproduces the leading term of the recurrence, and it is what forces a harmonic decomposition to exist at all.

Theorem 6.19 (Franel). [PROVED] *modulo the certificate check described after Theorem 6.20; the displayed identities are [PROVED] (**Factor** $\rightarrow 0$, exact). With $A(n) = \sum_k \binom{n}{k}^3$ (A000172), $(n+1)^2 u_{n+1} = (7n^2 + 7n + 2)u_n + 8n^2 u_{n-1}$ and $B(n) = \sum_k \binom{n}{k}^3 [\frac{1}{8}(H_k^{(2)} + H_{n-k}^{(2)}) + \frac{3}{8}(H_k - H_{n-k})^2]$, one has $(LB) = 0$ for the Franel operator, so B is the second solution, with values $B(0), \dots, B(8)$ equal to 0, 1, 4, $\frac{208}{9}$, $\frac{1280}{9}$, $\frac{208384}{225}$, $\frac{1404928}{225}$, $\frac{95174656}{2205}$, $\frac{3351248896}{11025}$. The certificates are*

$$\rho_F(n, k) = -\frac{1}{n}(4 - 12k + 12k^2 - 4k^3 + 22n - 39kn + 18k^2n + 32n^2 - 27kn^2 + 14n^3),$$

$$G_F = \frac{k^3 \rho_F \binom{n+1}{k}^3}{(n+1)^3},$$

$z_1 = N_F(k)/(n+1-k)^4$, $z_0 = M_F(k)/(n+1-k)^5$ with

$$N_F = \frac{3k^2}{4n}(4 - 16k + 24k^2 - 16k^3 + 4k^4 + 26n - 68kn + 63k^2n - 20k^3n \\ + 54n^2 - 88kn^2 + 39k^2n^2 + 46n^3 - 36kn^3 + 14n^4),$$

$$M_F = -\frac{k}{2n}(2 - 10k + 20k^2 - 20k^3 + 10k^4 - 2k^5 + 15n - 50kn + 70k^2n - 45k^3n + 11k^4n \\ + 40n^2 - 90kn^2 + 80k^2n^2 - 25k^3n^2 + 50n^3 - 70kn^3 + 30k^2n^3 \\ + 30n^4 - 20kn^4 + 7n^5),$$

and the boundary identities, all **Factor** $\rightarrow 0$, are $\rho_F(n, n+1) = -(n+1)^2$, $N_F(n) = \frac{3}{4}((n+1)^5 - (n+1))$, $M_F(n) = -\frac{1}{2}(1 + (n+1)^5)$.

Theorem 6.20 ($s_{10} = \sum_k \binom{n}{k}^4$). [PROVED] *modulo the certificate check described below. With $(n+1)^3 u_{n+1} = (2n+1)(6n^2 + 6n + 2)u_n + n(64n^2 - 4)u_{n-1}$ and $B(n) = \sum_k \binom{n}{k}^4 [\frac{1}{10}(H_k^{(2)} + H_{n-k}^{(2)}) + \frac{2}{5}(H_k - H_{n-k})^2]$, one has $(LB) = 0$, with $B(0), \dots, B(6)$ equal to 0, 1, $\frac{21}{4}$, $\frac{1001}{18}$, $\frac{85085}{144}$, $\frac{4203199}{600}$, $\frac{52055003}{600}$. The certificate is $G_{10} = k^4 \rho_{10} \binom{n+1}{k}^4 / (n+1)^4$ with*

$$\rho_{10}(n, k) = -\frac{1}{n^3}(-4k + 16k^2 - 24k^3 + 16k^4 - 4k^5 \\ + 10n - 72kn + 172k^2n - 184k^3n + 90k^4n - 16k^5n \\ + 80n^2 - 368kn^2 + 600k^2n^2 - 416k^3n^2 + 104k^4n^2 \\ + 255n^3 - 796kn^3 + 818k^2n^3 - 276k^3n^3 \\ + 385n^4 - 756kn^4 + 374k^2n^4 + 275n^5 - 260kn^5 + 75n^6),$$

$z_1 = N_{10}(k)/(n+1-k)^5$ and $z_0 = M_{10}(k)/(n+1-k)^6$ (both of degree 9 in k , with factors k^3 and k^2 ; full coefficients in the certificate file, see Section 10), and boundary identities $\rho_{10}(n, n+1) = -(n+1)^3$, $N_{10}(n) = \frac{4}{5}((n+1)^7 - (n+1)^2)$, $M_{10}(n) = -\frac{1}{2}((n+1) + (n+1)^7)$, all **Factor** $\rightarrow 0$.

Remark 6.21 (what the two theorems above do and do not contain). Both are proved by the same route as Theorem 6.16: one exhibits ρ , G , z_1 , z_0 , checks $\Delta_k G = \rho S$ and the two Gosper identities as polynomial identities in $\mathbb{Q}(n)[k]$, and checks the boundary identities. Every displayed identity above was verified that way, by exact expansion to 0. What is *not* written out here is the derivation: we do not state the Gosper key equations that z_1, z_0 solve, and for Theorem 6.20 the polynomials N_{10} , M_{10} themselves are not printed — their full coefficients are in the certificate file (Section 10). A reader wanting a self-contained proof must read that file. We flag this because [PROVED] is defined in Section 1.4 as “a complete proof is written out here”, and for these two theorems the certificate half is delegated rather than printed.

Observation 6.22 (the pattern). With $\beta_d = \frac{d}{2(d+1)}$ the δ^2 -coefficient and $c_+(n) = (n+1)^{d-1}$, both theorems read

$$\begin{aligned}\rho(n, n+1) &= -(n+1)^{d-1}, & N(n) &= 2\beta_d((n+1)^{2d-1} - (n+1)^{d-2}), \\ M(n) &= -\frac{1}{2}((n+1)^{2d-1} + (n+1)^{d-3}).\end{aligned}$$

It is natural to conjecture that this is the general- d shape, i.e. that Chamberland–Straub’s conjecture that $A^{(d)}$ has Apéry limit $\zeta(2)/(d+1)$ is provable this way for every d for which the sequence satisfies an order-two recurrence [VERIFIED: $d = 3, 4$ only].

6.6. A sharp obstruction: the $\zeta(2)$ -Apéry case. The template of the previous subsection does not merely fail to apply to the $\zeta(2)$ -Apéry sequence $\mathbf{D}$; it provably cannot apply.

Let $A(n) = \sum_k \binom{n}{k}^2 \binom{n+k}{k}$ (A005258: 1, 3, 19, 147, 1251, ...), $(n+1)^2 u_{n+1} = (11n^2 + 11n + 3)u_n + n^2 u_{n-1}$, and $B(n) = \sum_k \binom{n}{k}^2 \binom{n+k}{k} \cdot \frac{1}{5} [H_n^{(2)} + H_k(2H_k - H_{n-k} - H_n)]$, with $(LB) = 0$ [VERIFIED: $n \leq 25$]. Everything up to the Gosper step goes through, and beautifully: the Zeilberger certificate is the smallest in this paper,

$$\rho_{\mathbf{D}}(n, k) = k^2 + k(1 + 6n) - 4 - 15n - 11n^2, \quad G_{\mathbf{D}}(n, k) = \frac{k^3 \rho_{\mathbf{D}}(n, k) \binom{n+1}{k}^2 \binom{n+k-1}{k}}{n(n+1)^2},$$

[PROVED] (Expand $\rightarrow 0$; [VERIFIED: pointwise, $n = 1..9$, $k = 0..n+3$]), and the structural boundary identity (b1) holds exactly:

$$\rho_{\mathbf{D}}(n, n+1) = -2(n+1)(2n+1) = -(n+1)^2 \frac{\binom{2n+2}{n+1}}{\binom{2n}{n}} = -c_+(n) \frac{S(n+1, n+1)}{S(n, n)}.$$

The correction decomposes as $Y(n, k)/S(n, k) = A + B_1 H_k + B_2 (H_{n-k} + H_n)$ with $B_2 = \rho_{\mathbf{D}}(n, k+1)/(5(k+1))$ [VERIFIED: exactly, $n = 2..8$, $0 \leq k \leq n-1$].

Proposition 6.23 (the H_{n-k} component is not Gosper-summable). [PROVED] Let $T(k) := S_{\mathbf{D}}(n, k) B_2(n, k)$. In Gosper’s decomposition $T(k+1)/T(k) = \frac{a(k)}{b(k)} \frac{c(k+1)}{c(k)}$ one has $a = (n-k)^2(n+k+1)$, $b = (k+1)^2(k+2)$, $c(k) = \rho_{\mathbf{D}}(n, k+1)$, and $\gcd(a(k), b(k+j)) = 1$ for all $j \geq 0$ (the roots are $n, -n-1$ versus $-j-1, -j-2$, with n generic). Gosper’s key equation is $a(k)x(k+1) - b(k-1)x(k) = c(k)$. Since

$$a(k) - b(k-1) = -n(k^2 + (n+2)k - n(n+1))$$

has degree 2 with leading coefficient $-n \neq 0$, the degree bound forces $\deg x = 0$; and $x = x_0$ constant forces $x_0 = -1/n$ from the k^2 coefficient, which then produces the k -coefficient $n+2$ against the required $3+6n$. Contradiction. Independently [VERIFIED]: the corresponding linear system has no solution for any $\deg x \leq 8$.

Remark 6.24 (diagnosis). In the Franel and s_{10} cases the weight is $k \leftrightarrow n - k$ symmetric, the certificate module has rank 2, and each component telescopes. Here the weight is genuinely three-lettered (H_k, H_{n-k}, H_n) , the components do not telescope separately, and the weight-two part of any enlarged ansatz is forced to vanish ($\Delta(Sz) = 0 \Rightarrow z = 0$). Closing this case needs either a different gauge for w — the harmonic-monomial fit has a large kernel, 11-dimensional in the $\zeta(3)$ case, and a kernel shift changes the antidifference — or a genuine $\Pi\Sigma$ -field ansatz allowing nested sums rather than rational $\times$ harmonic-monomial. This is [OPEN], and it is the first place in this programme where the rank-two mechanism is provably insufficient.

6.7. Literature. The recurrence $(Lb) = 0$ itself is not new. It is proved in full by Schneider [27] using Karr’s $\Pi\Sigma$ difference fields and creative telescoping, with a certificate whose numerator, after the substitution $n \mapsto n - 1$, is exactly the numerator of our $G(n, k)$ — an independent confirmation of the normalisation of (6.3). Historically it goes back to the Zagier–Cohen reorganisation of Apéry’s proof reported by van der Poorten [23, §8], who gives $B_{n,k} = 4(2n+1)(k(2k+1) - (2n+1)^2) \binom{n}{k}^2 \binom{n+k}{k}^2$ and, for the second solution, $A_{n,k} = B_{n,k}c_{n,k} + [5(2n+1)(-1)^{k-1}k/(n(n+1))] \binom{n}{k} \binom{n+k}{k}$ — precisely our Step-D certificate, with $G(n, k) = B_{n,k-1}$ and V as in Lemma 6.14. But van der Poorten dispatches the verification as “after some massive reorganisation (9) becomes $A_{n,k} - A_{n,k-1}$ ”: it is exhibited, not written out; Fischler [26, §1.2] likewise calls it “une simple vérification”. Zudilin [21, eqs. (5)–(6)] has the rational-function form $s_n(t) = 4(2n+1)(-2t^2 + t + (2n+1)^2)$, the same polynomial under $t \leftrightarrow -k$.

What appears to be new is the *minimal form* itself. We could not locate $2H_n^{(3)} - H_k^{(3)}$ anywhere: not in OEIS A059415 (which is exactly this b_n , and whose only recorded closed form is the classical double sum), nor in [7, 11, 17, 8], nor in [23, 27, 26, 25]. The nearest published relative is Nesterenko’s Lemma 1 [24], reproduced as formula (3) of [21], which amounts to the weight $H_k^{(3)} + (2H_k - H_{n+k} - H_{n-k})H_k^{(2)}$ — a longer representation, not the minimal one. (The harmonic-monomial fitting system for b_n has an 11-dimensional kernel, so such representations are far from unique; the pivot choice delivered the shortest.) Similarly, (D-BIN) of Lemma 6.15 we could not locate as a named identity, though it is one line from Chu–Vandermonde: from $\sum_{k \geq 0} (-1)^k \binom{n+1}{k} \binom{n+k-1}{k} x^k = {}_2F_1(-(n+1), n; 1; x)$, apply $x d/dx$ at $x = 1$ to get $-n(n+1) {}_2F_1(-n, n+1; 2; 1) = 0$ for $n \geq 1$, and peel off the $k = n+1$ term. Its relative $\sum_k (-1)^k \binom{n}{k} \binom{n+k}{k} = (-1)^n$ is $P_n(-1)$. Finally, Paule and Schneider [25] prove Ahlgren’s $\sum_j (1 - 4jH_j + 4jH_{n-j}) \binom{n}{j}^4 = (-1)^n \binom{2n}{n}$ — the same right-hand side and the same $d = 4$ summand as Theorem 6.20 — but only weight-one identities; $H^{(3)}$ never appears there, so Theorem 6.20 is not covered by that work.

7. SHARPNESS OF THE HYPOTHESIS $p \geq 5$

Every statement in this paper carries the hypothesis $p \geq 5$. This section records where that hypothesis is used, where it can be removed, and what actually happens at $p = 2, 3$.

7.1. Where $p \geq 5$ enters the proofs. In Theorem 2.18 the hypothesis is used twice, and only twice: the coefficient $1/(2m^3)$ of Apéry’s letter $u(n, m)$ requires $p \nmid 2$, and the identity $(-1)^{jp-1} = (-1)^{j-1}$ in the proof of Lemma 2.16 requires p odd. Nothing else in Section 2 sees the size of p ; in particular Theorem 2.11 is uniform in p .

In Theorem 4.4 the hypothesis enters through (H4), in the requirement $c_j \in \mathbb{Z}_{(p)}$ on the coefficients of the weight. This is not a formality: it is the only obstruction the mechanism ever meets in the tame cases, and it bites exactly where the denominators say it should. For $\mathbf{D}$, whose coefficients are $\frac{1}{5}(1, 2, -1, -1)$, the two-layer split has 11 failing cells at $p = 5$ and none

at $p = 7, 11, 13$ [VERIFIED]; for s_7 , whose coefficients have denominator 28, it has 13 failing cells at $p = 7$ and none at $p = 5, 11, 13$ [VERIFIED]. The proof is therefore unconditional for $p \geq 7$ in those two cases, and $p = 5$ (resp. $p = 7$) is [VERIFIED] only. Curiously s_{10} , whose coefficients are also $\frac{1}{5}(1, 4, -4)$, exhibits no defect at $p = 5$ — the layer valuations happen to be one higher there — but the proof as written still requires $p \geq 7$.

7.2. Where it can be removed. The minimal form of Theorem 6.1 has coefficients 2 and -1 , both integers, so (H4)'s coefficient condition becomes vacuous. Theorem 4.4 is stated for $p \geq 5$, but that hypothesis is used *only* through (H4): inspecting its proof, every other step is uniform in p . So the proof — not the theorem as stated — yields the γ instance for *every* prime, including 2 and 3. This is machine-verified: the Lean statement of Section 9 carries no hypothesis $p \geq 5$ and no hypothesis $a \geq 1$. The same inspection gives the Franel instance for every $p \neq 2$ (its coefficient denominators are 4), so that the $p \geq 5$ recorded for **A** in Section 5 is not sharp: $p = 3$ is fine.

So the exclusion of small primes is a property of the *normalisation of the weight*, not of the congruence. In the natural normalisation $B(0) = 0$, $B(1) = 1$ the γ coefficients are $\frac{1}{3}, \frac{1}{6}$ and $p \geq 5$ is required; in the classical normalisation $b_n = 6B(n)$ they are 2, -1 and it is not.

7.3. What actually happens at $p = 2$ and $p = 3$. The master law itself, however, does fail at the small primes, and not only for normalisation reasons.

Observation 7.1. [VERIFIED] With the twisted law of Conjecture 3.6 and $n \leq 120$, the floor drops below w for

$$\begin{aligned} \mathbf{A} \ (p = 2, \text{ floor } 1), \quad \mathbf{F} \ (p = 2, 1), \quad \alpha \ (p = 2, 2; p = 3, 2), \\ \gamma \ (p = 3, 2), \quad \delta \ (p = 3, 1), \quad \varepsilon \ (p = 2, 2). \end{aligned}$$

Independently, the iterated two-digit form $p^{2w}b_n \equiv b_{n_2}a_{n_1}a_{n_0}$ holds at $p = 5$, $w = 3$, $n = 25..70$ with minimum depth 1, and fails at $p = 2$ and $p = 3$ with minimum orders -6 and 0 respectively.

Six of the fifteen families therefore violate the master floor at 2 or at 3, and the violations are of the same kind as the low-weight ratio failures of Section 3.4: the floor does not merely dip, it drops by a full unit or more. The boundary $p \geq 5$ that this forces coincides with the boundary already familiar from the denominator-conjecture side of this programme, where the primes 2 and 3 are likewise excluded.

We record honestly that we have *no* conceptual explanation of the $p = 2, 3$ failures. The proof-side obstruction (coefficient denominators) accounts for individual cells, and it is removable by renormalisation; the observed collapse of the master floor at 2 and 3 for six families is a separate phenomenon, and it is [OPEN].

8. DISCUSSION

8.1. Relation to Beukers–Vlasenko’s Conjecture 7.5. The natural home for a congruence like Conjecture 3.6 is the theory of Dwork crystals, and it is worth being precise about how the two fit together. We read [4] from its source; its §7 treats *completely symmetric Calabi–Yau families*, and its numbered items there are 7.1 (definition), 7.2 (lemma), 7.3 (theorem), 7.4 (remark) and 7.5 (conjecture).

The setting is $f(x) = 1 - tg(x)$ and $F(t) = \sum_n f_n t^n$ with $f_n = \text{ct}(g(x)^n)$; for the sporadic families these f_n are exactly our $A(n)$. Writing $F_N(t)$ for the truncation of F at t^N , and $G(t)$ for the series appearing in the second solution $F(t) \log t + G(t)$ of the (second-order) Picard–Fuchs

operator, *the coefficients of G are our $B(n)$* , up to normalisation. What [4] proves — their eq. (7.6), from [3, eq. (7)] — is that for all $m, s \geq 1$,

$$\frac{F(t)}{F(t^\sigma)} \equiv \frac{F_{mp^s}(t)}{F_{mp^{s-1}}(t^\sigma)} \pmod{p^s}. \quad (8.1)$$

Reading off coefficients with the naive lift $t^\sigma = t^p$ and $s = 1$ gives the Dwork–Mellit–Vlasenko congruence whose sporadic specialisation is precisely the first-row Lucas congruence $A(ap+r) \equiv A(a)A(r) \pmod{p}$ of Theorem 2.11. Their Conjecture 7.5 asserts that (8.1) holds modulo p^{2s} — for $m = 2$ in the hypercubic and hyperoctahedral families and $m = n + 1$ in the simplicial ones — where σ is the *excellent* Frobenius lift, the unique lift with $q^\sigma = \gamma^{p-1}q^p$ for the canonical (mirror) coordinate $q(t) = t \exp(G(t)/F(t))$; by their Theorem 7.3 that lift is characterised by the vanishing of the off-diagonal Frobenius entry. The authors state that they could not prove 7.5 for a single g .

Four points relate this to Conjecture 3.6.

- (1) *They are the two rows of one 2×2 Frobenius matrix.* The crystal $\text{CY}(g)(2)$ of [4] is free of rank two on $1/f$, $\theta(1/f)$, and its Frobenius has diagonal (λ_0, λ_2) with $\lambda_0 = F(t)/F(t^\sigma)$ the unit root. Conjecture 7.5 is a statement about λ_0 alone, sharpened from modulo p^s to modulo p^{2s} . Conjecture 3.6 is the statement about the *other* diagonal entry: that Frobenius acts on the second graded piece with eigenvalue $\chi(p)p^w$. In coefficient form, their λ_0 -congruence *is* the first row; ours *is* the second row. Neither formally implies the other.
- (2) *But 7.5’s standing hypothesis is a weak form of our conclusion.* The excellent lift exists as a lift of $\mathbb{Z}_p[[t]]$ precisely because $q(t) = t \exp(G/F) \in \mathbb{Z}_p[[t]]$ — p -integrality of the mirror map, which is a statement about G , i.e. about $B(n)$. Our Corollary 4.8, $d_n^w B_n \in \mathbb{Z}_{(p)}$, obtained by iterating the descent over base- p digits, is exactly the coefficient-level shadow of that integrality, and it is sharper: it does not merely assert integrality of the mirror map, it pins the depth at exactly w and exhibits the eigenvalue $\chi(p)$. So the implication runs

$$\begin{aligned} \text{Conjecture 3.6} &\implies (\text{coefficient form of mirror-map integrality}) \\ &\implies (\text{the excellent lift is defined over } \mathbb{Z}_p[[t]]), \end{aligned}$$

which is the standing hypothesis of Conjecture 7.5. In this precise sense our conjecture is *upstream* of theirs, and not equivalent to it.

- (3) *The character is invisible in the unit-root formulation and must be added.* Beukers and Vlasenko work with $\lambda_0 = F/F^\sigma$, where no character can appear: the unit root of the holomorphic period is a unit, and the first-row Lucas congruence is untwisted for all fifteen families — [VERIFIED] here and proved in [13]. The character surfaces only on the *non-unit* eigenvalue. Any crystal-theoretic proof of Conjecture 3.6 must therefore carry the information that, for seven of the fifteen families, the second Frobenius eigenvalue is $\chi(p)p^w$ and not p^w — i.e. that the family is attached to a quadratically twisted (CM) modular parametrisation. This is a concrete correction to make to any slogan of the form “Dwork crystals imply Lucas congruences for the second solution”.
- (4) *What our conjecture does not give.* Nothing towards the modulo- p^{2s} doubling that is the actual content of 7.5. Our floor is exactly w and provably not more in the sense that equality is attained in every cell tested, so there is no hidden supercongruence in the second row of the kind 7.5 predicts for the first.

We note for completeness that we could not find any second-solution version of these congruences stated in [4] or in Vargas-Montoya’s work [20, 19]; [19] proves Lucas congruences explicitly and exclusively for the holomorphic solution, and [20] proves p -integrality of $\exp(y_1/y_0)$ with no congruence attached.

8.2. Frobenius, not the archimedean limit. Observation 5.9 deserves emphasis. Three sporadic sequences — $\mathbf{B}$, δ , η — have characteristic roots forming a complex-conjugate pair of equal modulus, so Poincaré–Perron gives no dominant solution and $B(n)/A(n)$ oscillates forever: there is no Apéry limit to speak of. Nonetheless all three satisfy Conjecture 3.6, with a flat floor, and with characters χ_{-3} , trivial and χ_5 respectively.

The p -adic Apéry limit therefore exists where the archimedean one does not. This is the sharpest indication available to us that the descent is a statement about a Frobenius eigenvalue and not about an asymptotic ratio: the archimedean limit is an accident of the relative sizes of the characteristic roots, whereas the p -adic descent is insensitive to them. It also means that “the character of the Apéry limit” — which is how we identified χ in twelve of the fifteen cases — cannot be the definition of χ ; for those three it must be read off from the descent itself, or from the modular parametrisation.

8.3. Beyond rank two. The natural next target is the Brown–Zudilin cellular pair for $\zeta(5)$ [6], where the integral row is a double binomial sum Q_n , the second and third solutions P_n and $\widehat{P}_n$ satisfy the same order-three recurrence, and the expected descent has $w = 5$. The obstructions are structural and were identified early in this programme: the local system has rank three, so the descent must be proved for a three-step filtration at once, with the intermediate weight-three object $\widehat{P}_n$ appearing in the P_n descent as a cross-term with no $\zeta(3)$ analogue; the summand is not a perfect square, so the per-carry valuation slack of Remark 2.15 disappears; and the pole to be assembled is of order five rather than three, which roughly squares the case analysis of Lemma 2.14. The first-row Lucas congruence $Q_{ap+r} \equiv Q_a Q_r \pmod{p}$ is proved and formalised (Section 9).

The lesson of Section 6 applies here with force: the right search is not “which weight-five harmonic monomials span P_n ”, but “which weight has an n -difference equal to a single hypergeometric term times a rational function” — a smaller, linear and decidable search, and one insensitive to the large kernel that makes monomial fitting degenerate. And the top-cell constant, which at weight three is the classical 5 of Lemma 6.13, is computable in advance from the residue at the top of the support, giving a necessary numerical fingerprint that any candidate weight must match.

8.4. Open problems.

Problem 8.1. Prove Conjecture 3.6, in any of its three independent directions: the multi-digit statement, the sharpening from modulo p to modulo p^w , and the removal of the tameness hypothesis. Even the weight-three case, Conjecture 3.1, is open in all three.

Problem 8.2. Prove Theorem 2.18 by an ε -deformation. Gessel’s companion congruence [1, Thm. 1.4] is structurally the first parameter-derivative of the first-row congruence; our b_n is the weight-three companion, i.e. the third derivative. Writing $a_n = \text{ct}[\Lambda_\varepsilon^n]$ for a deformed Laurent polynomial, applying the Dwork–Mellit–Vlasenko ratio congruence [14] to the deformed family and taking $\partial_\varepsilon^3|_{\varepsilon=0}$ should land the weight-three pole and multiply the modulus by p^3 , since each filtration level costs one power of p [4, Prop. 7.7].

Problem 8.3. Supply the valuation bound (“a Kummer carry in a wide binomial beats the order-one pole of a harmonic letter at $\lfloor 2k/p \rfloor$ ”) that would remove the tameness hypothesis (H4) and put α , ε , s_7 and $\mathbf{E}$ inside Theorem 4.4. This would also produce the first *proved* instance of the twisted case $e = 1$, which at present has no tame example.

Problem 8.4. Find harmonic decompositions for the remaining seven families. For $\mathbf{C}$ this provably requires letters that its own summand does not supply (Observation 5.8); a constant-term representation [11] is the natural route. For δ , letters attached to the 3-section of the summand.

Problem 8.5. Close the $\zeta(2)$ -Apéry case of Section 6.6: either exhibit a gauge (a shift by an element of the 11-dimensional kernel of the fitting system) in which the correction telescopes, or run the argument in a $\Pi\Sigma$ field with nested sums. Proposition 6.23 shows no rational $\times$ harmonic-monomial ansatz can work.

Problem 8.6. Give an intrinsic definition of χ — one that does not go through the Apéry limit, and so applies to $\mathbf{B}$, δ , η — and prove that it is the character of the quadratic twist of the modular parametrisation attached to the family.

9. MACHINE VERIFICATION

A substantial part of this paper is formalised in Lean 4 and checked by its kernel. This section reports exactly what is and is not machine-verified. Toolchain: `leanprover/lean4:v4.33.0-rc1`, Mathlib revision `cd580e54f1a6b46063824e80cec92f64692cbe78`.

9.1. What is formalised.

result	status
Theorem 2.11 (first row, Apéry)	formalised, 0 sorries
first row for the Brown–Zudilin double sum Q_n	formalised, 0 sorries
full digit-product form of both first rows	formalised, 0 sorries
Lemma 4.1 and Corollary 4.2	formalised, 0 sorries
Theorem 4.4 (general χ -twisted descent)	formalised, 0 sorries
second row for the minimal form $B_{\min}$ (the $B_{\min}$ analogue of Theorem 2.18)	formalised, 0 sorries
second row for Franel	formalised, 0 sorries
two- and three-digit Kummer module (K1–K4)	formalised, 0 sorries
$B_{\min} = b$ (Theorem 6.1)	not formalised (proved on paper)
Conjecture 3.6 (master form)	not formalised (not proved)

`lake build` is clean; a clean rebuild of all eight modules takes 34 seconds with Mathlib prebuilt. A `grep` for `sorry`, `admit` and `native_decide` over the development returns nothing, and `#print axioms` on every exported theorem returns only the three standard Mathlib axioms:

```
'ZetaLucas.K_descent'      depends on axioms:
    [propext, Classical.choice, Quot.sound]
'ZetaLucas.mon_descent'    depends on axioms: [ ... same three ... ]
'ZetaLucas.theorem_LB'     depends on axioms: [ ... same three ... ]
'ZetaLucas.bMin_lucas'     depends on axioms: [ ... same three ... ]
'ZetaLucas.bFranel_lucas' depends on axioms: [ ... same three ... ]
'ZetaLucas.apery_lucas'    depends on axioms: [ ... same three ... ]
'ZetaLucas.Q_lucas'       depends on axioms: [ ... same three ... ]
```

In particular there is no `sorryAx` and no `Lean.ofReduceBool`: no statement rests on a compiled-code evaluation.

9.2. The statements. We display the main statements in ASCII-transliterated Lean (Unicode identifiers such as $\mathbb{Q}$, $\mathbb{N}$, $\in$, $\forall$ are written `Q`, `N`, `in`, `forall`). The predicate `PCong p x y` is defined as `padicNorm p (x - y) < 1`, i.e. $x \equiv y \pmod{p}$ inside $\mathbb{Z}_{(p)} \subseteq \mathbb{Q}$; it implies p -integrality of both sides as soon as one of them is p -integral, so the statements are not vacuous.

```
-- Lemma K
theorem K_descent {chi : N -> Z} (hchi : forall m n, chi (m*n) = chi m * chi n)
  {r : N} (hr : 0 < r) (y : N) :
  PCong p ((p : Q)^r * K chi r y) ((chi p : Q) * K chi r (y / p))

-- Theorem LB (hypotheses H1, H2, H3, H4, H4c, Hw, H5 as in Section 4)
theorem theorem_LB {J : Finset iota} {c : iota -> Q}
  {mon : iota -> List Letter} {S : N -> N -> Z} {w : N} {chip : Z}
  (H1 ...) (H2 ...) (H3 ...) (H4 ...) (H4c ...) (Hw ...) (H5 ...)
  {a r : N} (ha : a < p) (hr : r < p) :
  PCong p ((p : Q)^w * Brow J c mon S (a*p + r))
    ((chip : Q) * Brow J c mon S a * (Arow S r : Q))

-- the second row for the minimal Apéry form; bMin n is the sum over
-- k in range (n+1) of (A n k : Q) * (2 * Harm 3 n - Harm 3 k)
theorem bMin_lucas {p : N} [Fact p.Prime] {a r : N}
  (ha : a < p) (hr : r < p) :
  PCong p ((p : Q)^3 * bMin (a*p + r)) (bMin a * (apery r : Q))

-- the second row for Franel
theorem bFranel_lucas {p : N} [Fact p.Prime] (hp2 : p <> 2) {a r : N}
  (ha : a < p) (hr : r < p) :
  PCong p ((p : Q)^2 * bFranel (a*p + r)) (bFranel a * (franel r : Q))
```

9.3. Two sharpenings found by formalising.

- (1) `bMin_lucas` carries *no* hypothesis $p \geq 5$ and *no* hypothesis $1 \leq a$. The minimal weight's coefficients are the integers $2, -1$, so the (H4) coefficient condition is vacuous and the theorem holds for every prime, including 2 and 3; and $a = 0$ is fine, since $B_{\min}(0) = 0$ and $p^3 B_{\min}(r)$ is p -divisible for $r < p$. The informal Theorem 2.18 needs $p \geq 5$ only because Apéry's original weight carries $1/(2m^3)$.
- (2) `bFranel_lucas` needs only $p \neq 2$, where the informal analysis had recorded $p \geq 5$; the coefficient denominators are 4, and $p = 3$ is fine.

9.4. Design decisions, and why they matter mathematically. Three changes were made to the informal hypotheses; all three make the theorem easier to instantiate, not harder.

- (1) (H1) is stated as an unconditional congruence $S(ap + r, bp + s) \equiv S(a, b)S(r, s) \pmod{p}$ for *all* a, b , together with $S(n, k) = 0$ for $k > n$; (H2) then disappears. In the carrying regime both sides vanish modulo p , so the dichotomy is absorbed. The consequence is that *no product-region argument occurs anywhere in the formalisation*: the double sum factors because a full block $\{0 \leq k < (a + 1)p\}$ splits, and the extension from $\{0 \leq k \leq n\}$ to that block is an equality of integers. This retires, for the general theorem, the compression that Lemma 2.10 had to expand by hand.
- (2) (H3)'s "surviving set" is replaced by the side condition $S(r, s) \not\equiv 0 \pmod{p}$, which is exactly what excludes a borrow in an argument such as $n - k$: for Franel it gives $\binom{r}{s} \neq 0$, hence

$s \leq r$, hence $\lfloor (n-k)/p \rfloor = a-b$. Instances then *derive* the no-borrow condition rather than describing Σ_r .

- (3) (H5) is stated as “the product of the letters’ characters at p is the same for every monomial” rather than as a count of χ -letters. This is strictly more general — it allows mixed conductors — and cheaper to check.

The letter system is deliberately concrete: a **Letter** bundles a completely multiplicative $\chi : \mathbb{N} \rightarrow \mathbb{Z}$, a positive degree, and an argument form $\mathbb{N} \rightarrow \mathbb{N} \rightarrow \mathbb{N}$; a monomial is a **List Letter**, so genuine products such as $H_k H_{n-k}$ (which Franel needs) are expressible. The sum $K_\chi^{(r)}(y)$ is taken over **range** $(y+1)$, so the $m=0$ term is $\chi(0)/0^r = 0$ in $\mathbb{Q}$ for $r \geq 1$; this makes the reindexing of the p -divisible layer a single bijection $m \mapsto m/p$, $j \mapsto jp$, and no reindexing of **Finset.Icc** — the step that had been priced as the main schedule risk — ever arises.

9.5. Numerical checks inside the build. Normalisations are pinned in-file, by kernel evaluation where feasible and by **#eval** otherwise: **apery** $\mapsto 1, 5, 73, 1445, 33001, 819005$ (A005259, by **decide**); $Q \mapsto 1, 21, 2989, 714549, 217515501, 76157194521$, confirming the double-sum reading $Q_1 = 21$; **bMin** $\mapsto 0, 6, \frac{351}{4}, \frac{62531}{36}, \frac{11424695}{288}, \frac{35441662103}{36000}$, which is exactly the classical sequence b_n ; **franel** $\mapsto 1, 2, 10, 56, 346, 2252$ (A000172) and **bFranel** $\mapsto 0, 1, 4, \frac{208}{9}, \frac{1280}{9}$, matching the Franel recurrence with $B(0) = 0, B(1) = 1$. An evaluable form of p -divisibility sweeps every cell: both second-row congruences evaluate to **true** for all $a, r < p$ at $p = 5, 7$ in the build, and off-line at $p = 11, 13$. Sharpness spot-check: at $p = 5, a = r = 1$ the difference $p^3 B_{\min}(ap+r) - B_{\min}(a) a_r$ equals $103188530395/32$, of 5-valuation exactly 1 — the floor is exactly w , as the sweeps of Section 3 report.

9.6. Effort, and the economics of the two routes. The development is 1540 lines across eight modules (**Core**, **Apéry**, **BrownZudilin**, **PadicBridge**, **Letters**, **TheoremLB**, **Instances**, **Kummer**).

The route matters more than the rate. Formalising Theorem 2.18 *via the proof of Section 2* — Lemma 2.14, Corollary 2.12, Lemma 2.9, the Kummer ledger and the block reindexing of **Finset.Icc** — was costed in advance at ≈ 13 hours ± 4 , with two pieces carrying essentially all the schedule risk. Formalising it *via Theorem 4.4 and the minimal form* cost ≈ 2.2 hours and produced a strictly stronger result: arbitrary χ , arbitrary weight w , all primes, $a = 0$ allowed, and a second sequence for free. The claim made informally — that the general theorem “reproves the weight-three theorem in three lines, with no Lemma V, no tail factorisation and no Kummer ledger” — is thus machine-confirmed, and the cost ratio is about $6\times$. (Wall-clock figures are reconstructed from the work order rather than stopwatch readings and should be read as $\pm 25\%$; line counts and build-iteration counts are exact.)

The Kummer module is retained even though the (LB_w^χ) route does not use it: it is precisely the toolkit that the non-tame instances of Section 5.5 will need.

9.7. What is *not* formalised.

- (1) $B_{\min} = b$. The Lean theorem is about the explicit sum $B_{\min}(n) = \sum_k A(n, k)(2H_n^{(3)} - H_k^{(3)})$. Its equality with the classical Apéry numerator is Theorem 6.1, proved on paper here and *not* formalised. The gap is therefore a formalisation gap, not a mathematical one, and it is well-scoped: Lemma 6.2’s six conversions, the six certificates of Table 4 (each an **Expand** $\rightarrow$ 0 polynomial identity, i.e. **ring** in Lean) and an induction on L , whose leading coefficient $(n+1)^3$ never vanishes.
- (2) $B_{\mathbf{A}}$ is the second Franel solution: same status — it means proving that the explicit sum satisfies the Franel recurrence, which is Theorem 6.19 here.

- (3) *The master form*, Conjecture 3.6: not proved, hence not formalised. Note that the first-row multi-digit form *is* formalised and is a cheap induction; the second row is not, because its ratio form has poles.
- (4) *A twisted instance*. The χ machinery of `theorem_LB` is proved but never exercised at $\chi(p) \neq 1$, because the only known twisted sequence (**E**) is non-tame (Problem 8.3).
- (5) *The non-tame instances and the seven families with no known decomposition*: nothing to formalise until the mathematics exists.

10. DATA AVAILABILITY, AND THE EXACT SWEEPS

Every computation reported in this paper is exact — rational arithmetic, or arithmetic in a finite field with rational reconstruction, or exact polynomial expansion. No floating point enters any claim, with the single exception of the PSLQ identification of the Apéry limits (Observation 5.1), which is a high-precision numerical identification and is labelled as such.

10.1. Scripts. The computations were carried out in the programme’s working repository, and the scripts are archived alongside this paper:

location	contents
<code>work/lbw/sporadic.py</code>	the fifteen sequences, their recurrences and second solutions
<code>work/lbw/t1b.py</code>	weights w , integrality, Apéry limits (PSLQ)
<code>work/lbw/t2_final.py</code> , <code>t2_diag.py</code> , <code>t2_twist.py</code>	the master sweeps, the failure diagnosis, the χ -twist
<code>work/lbw/t3_fit.py</code> , <code>t3_fit2.py</code>	harmonic-decomposition fitting and exact validation
<code>work/lbw/t4_proofcheck.py</code> , <code>work/lbw/deep.py</code>	per-layer checks of the proof of Theorem 4.4; depth data
<code>work/minform/step_a.wl</code> , <code>step_b.wl</code> , <code>step_b2.wl</code>	anchors and certificates for Theorem 6.1
<code>work/minform/certs_stretch.m</code> <code>lean/</code>	full certificates for Theorems 6.19 and 6.20 the Lean 4 development of Section 9 (eight modules)

10.2. The sweeps, with their exact parameters.

claim	range, and result
Theorem 2.11 (first row)	$5 \leq p \leq 31$, $n \leq 320$, single- and multi-digit: 0 failures
Theorem 2.18 (second row, single digit)	$p \in \{5, 7, \dots, 31\}$, all (a, r) : 0 failures; independently re-implemented for $p \in \{5, 7, 11, 13, 17\}$
Every lemma and case split of Section 2	$p \in \{5, 7, 11, 13\}$ (some to 19), all cells: 0 failures; $v_p(CC) = 3$ branch checked on all 14 292 occurrences for $p \leq 19$; the bound of Lemma 2.14 has minimum exactly 1
Conjecture 3.1 (master, $w = 3$)	$5 \leq p \leq 31$, $n \in [1, 320]$: 0 failures, floor exactly 3
Ratio-form depth on p -unit cells	$7 \leq p \leq 31$, single-digit: minimum exactly 3, no cell of depth ≤ 2
$v_p(b_n) \geq -3v_p(d_n)$	$5 \leq p \leq 31$, $n \leq 320$: 0 failures
Conjecture 3.6 (all fifteen)	every prime $5 \leq p \leq 103$, $n \leq 400$; plus $p = 5, 7, 11$ with $n \leq 1200$ (five base- p digits): 0 failures, floor exactly w in every cell
Failure of the untwisted law (Obs. 5.4)	$5 \leq p \leq 43$, $n \leq 300$
The repair (Obs. 5.5)	the seven twisted sequences, $5 \leq p \leq 31$ with $\chi(p) = -1$, $n < p^3$ (to 1500): valuation exactly w
Ramified prime (Obs. 5.6)	η , $p = 5$, $n \leq 500$: $\min_n v_5(B_n) = 0$
Small primes (Obs. 7.1)	$p = 2, 3$, $n \leq 120$
Weights and their sharpness (Obs. 5.2)	$n \leq 600$; $v_p(B_n) \geq -w \lfloor \log_p n \rfloor$ over $15 \times 16 \times 400$ cells
Apéry limits (Obs. 5.1)	PSLQ at 400 digits, 33-constant basis, agreement to 31–401 digits
Harmonic decompositions (Table 3)	fitted mod $2^{61} - 1$, rationally reconstructed, validated exactly over $\mathbb{Q}$ on held-out $n = 60..72$ ($n = 70..82$ for E)
Negative fits (Obs. 5.8)	C : bases of 14, 44, 44 and 152 elements against 110 and 220 exact equations; δ : 65 elements, 110 equations; E : pure bases up to 44 elements, 8 arguments
Layer-by-layer checks of Theorem 4.4	$\gamma, \mathbf{A}, s_{10}$ at $p = 5, 7, 11, 13$: 0 failures; D at $p = 7, 11, 13$: 0 failures, 11 failing cells at $p = 5$; s_7 : 13 failing cells at $p = 7$, none at $p = 5, 11, 13$
Certificates of Table 4	Expand $\rightarrow 0$; pointwise $n = 1..12$, $k = 0..n + 4$; $B_{\min} - b = 0$ exactly for $n \leq 50$ in two independent implementations
Theorems 6.19, 6.20	certificate identities pointwise $n = 1..10$ resp. $n = 1..9$; $(LB) = 0$ numerically for $n \leq 25$ resp. $n \leq 20$
Proposition 6.23	the Gosper linear system has no solution for any $\deg x \leq 8$

10.3. Provenance. The results were obtained in a sequence of exact computational campaigns, each recorded in a dated working memo, and each theorem in this paper was written by transcribing its statement from the memo that established it and re-checking it against that memo rather than reconstructing it from memory. Where a memo’s own numerical summary disagreed with the underlying sweep, or where a claim was later superseded, the correction is stated explicitly in the text: see Section 3.4 (the “dip at the reflection centre” that turned out to be a pole cell), Observation 3.2 and Observation 3.3 (two false strengthenings), and Section 5.5 (the primes at which the tame proofs actually apply).

ACKNOWLEDGEMENTS

This work was carried out in a human–AI collaboration: the conjectures were found by large exact computational sweeps and the proofs were drafted, refereed and formalised in a loop between the authors and language-model agents, with every statement re-checked against its source computation and every proof checked by hand or by machine.

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